

CHAPTER I

INTRODUCTION

Problem Background

The Sangguniang Kabataan (SK) plays a big role in supporting youth development and helping manage community activities at the barangay level. SK officials are responsible for planning programs, organizing events, and making sure that the needs of the youth are properly addressed. To carry out these responsibilities, they handle many important documents such as resolutions, activity proposals, accomplishment reports, meeting minutes, budgets, and approval forms. These documents serve as proof of transactions, references for decision-making, and records for accountability. Because of this, proper document management is very important for the smooth flow of SK operations.

However, in several barangays in Ormoc City specifically South, East, Ipil, Cagbuhangin, and Don Felipe Larrazabal most of these documents are still handled manually. Based on initial observations and informal conversations with SK officials, especially the SK Secretaries, documents are usually stored in folders, envelopes, cabinets, or sometimes even in boxes. While this method has been used for many years, it has become difficult to manage as the number of documents continues to grow.

In barangays like South and Cagbuhangin, where there are more activities and documents being produced, the lack of proper storage space makes it harder to keep records organized. Files are often stacked together, making it difficult to separate old and new documents. Over time, some documents get damaged due to moisture, dust,

pests, or simple wear and tear. There are also instances where documents are misplaced or cannot be found when needed. Because of this, SK officials spend a lot of time just searching for specific files, which slows down their work and affects productivity.

Another issue is the absence of a centralized system. In barangays such as East and Ipil, documents are sometimes kept in different locations or handled by different individuals. This makes it harder to monitor both ongoing and completed activities. Documents that require multiple approvals often experience delays because they need to be physically passed from one person to another. SK Secretaries shared that this process can be stressful, especially when deadlines need to be met. In addition, there is no reliable tracking system to check the status of documents, which can lead to confusion or duplication of work.

Security is also a concern. Since the documents are stored physically, they can be accessed by unauthorized individuals or accidentally lost without proper monitoring. Important and sensitive information may be exposed, which can affect the credibility and reliability of SK records. Without a proper system in place, it becomes difficult to ensure that documents are safe and well-maintained.

Aside from document-related concerns, managing SK member information is also challenging. In barangays like Don Felipe Larrazabal and Ipil, there is no centralized record system for tracking member status. Information about whether a member is active, inactive, or no longer participating is not always updated. This creates confusion when preparing reports, monitoring attendance, or verifying participation in activities. It also affects decision-making, since accurate data is not always readily available.

Overall, these challenges greatly affect the efficiency and effectiveness of SK operations. Instead of focusing more on developing programs and serving the youth, SK officials spend a significant amount of time dealing with paperwork and organizing records. This slows down processes, reduces productivity, and can even affect the quality of services provided to the community. With the continuous advancement of technology, relying solely on manual systems is no longer practical or efficient.

Because of this, the study proposes the development of a digital document management and monitoring system in coordination with the SK Chairman of Ormoc City. The proposed system will allow authorized users to securely store, organize, and access documents in a centralized platform. It will also provide features for tracking document status, monitoring SK activities, and managing member information more efficiently. Users will be able to view SK accounts, access member lists, and check their current status (active, inactive, or quit) in real time.

In addition, the system is expected to reduce the time spent on searching and retrieving documents, minimize errors, and improve overall organization. It will also enhance data security by limiting access to authorized users only. By digitizing records and processes, the system will make SK operations faster, more accurate, and more reliable.

In the long run, the proposed system aims to help SK officials focus more on their primary goal, which is to serve the youth and the community. It will support better decision-making, improve transparency and accountability, and contribute to a more organized and efficient SK governance.

Theoretical Background

The need for a more organized and systematic way of handling records becomes clear when looking at the current situation of the Sangguniang Kabataan (SK). As discussed in the introduction, the continued use of manual and paper-based processes makes it difficult to organize, retrieve, and monitor important documents. Because of these challenges, this study is anchored on Records Management Theory, which focuses on properly managing records throughout their lifecycle from creation to storage and eventual disposal. Applying this theory helps guide the development of the proposed system, SKActHub, which aims to improve how SK records in Ormoc City are organized, accessed, and monitored.

Records Management Theory

Records Management Theory explains that records should be handled in a structured and organized manner from the moment they are created up to the time they are no longer needed. This includes processes such as classification, storage, retrieval, and preservation. The theory views records as valuable assets because they serve as evidence of an organization's activities and decisions. According to the International Organization for Standardization (ISO 15489), effective records management ensures that records remain authentic, reliable, have integrity, and are usable over time. In simple terms, records should be trustworthy, protected from unauthorized changes, and easy to find when needed.

When records are not properly managed, several problems can arise. These include lost or duplicated documents, delays in decision-making, and difficulty in tracking important information. As explained by Shepherd and Yeo (2003), poor records management weakens institutional memory and reduces efficiency. In manual

systems, records are also more likely to be damaged, misplaced, or accessed by unauthorized individuals. As the number of documents increases, these problems become more serious and can slow down operations.

In the context of the SK, records such as resolutions, activity proposals, project plans, accomplishment reports, approvals, financial documents, and correspondence are essential. These documents serve as proof of activities and are needed for reporting, auditing, and transparency. However, because these are mostly stored physically, SK officials often experience difficulty in retrieving documents, monitoring activities, and verifying past decisions. This clearly shows the need for a more efficient system.

This is where Records Management Theory strongly supports the development of SKActHub. The proposed system follows the principles of the theory by digitizing records and organizing them in a structured way. Through features such as categorization, tagging, and indexing, SKActHub makes documents easier to find and manage. Instead of manually searching through folders, users can quickly retrieve files within seconds.

The theory also emphasizes the importance of managing the lifecycle of records. SKActHub supports this by allowing documents to move step-by-step from creation and submission to review, approval, storage, and monitoring. This means that documents are not just stored but are actively managed throughout their use. This directly solves the problem of scattered and untracked documents in the current system.

Another important principle of the theory is security and control. SKActHub applies this by implementing role-based access, where the SK Chairman can view all documents, while the Secretary and Treasurer can only access the files they uploaded or the folders they created. This ensures data confidentiality while still allowing

collaboration within the system. Backup features also help protect records from loss or damage.

Records Management Theory also supports transparency and accountability. With organized and accessible records, SK officials can easily prepare reports, respond to inquiries, and provide documentation when needed. This improves trust and ensures that all activities are properly recorded and traceable.

In addition, digital records management helps maintain continuity of operations. According to Penn, Pennix, and Coulson (1994), well-managed records ensure that organizations can continue functioning effectively even during changes in leadership. In the case of the SK, this means that new officers can easily access past documents, project histories, and reports. This prevents the loss of important information and supports smoother transitions.

Overall, Records Management Theory strongly supports this study because it provides the foundation for organizing, securing, and managing records efficiently. By applying this theory through SKActHub, the system not only addresses the current problems in document handling but also improves overall efficiency, transparency, and decision-making within the SK.

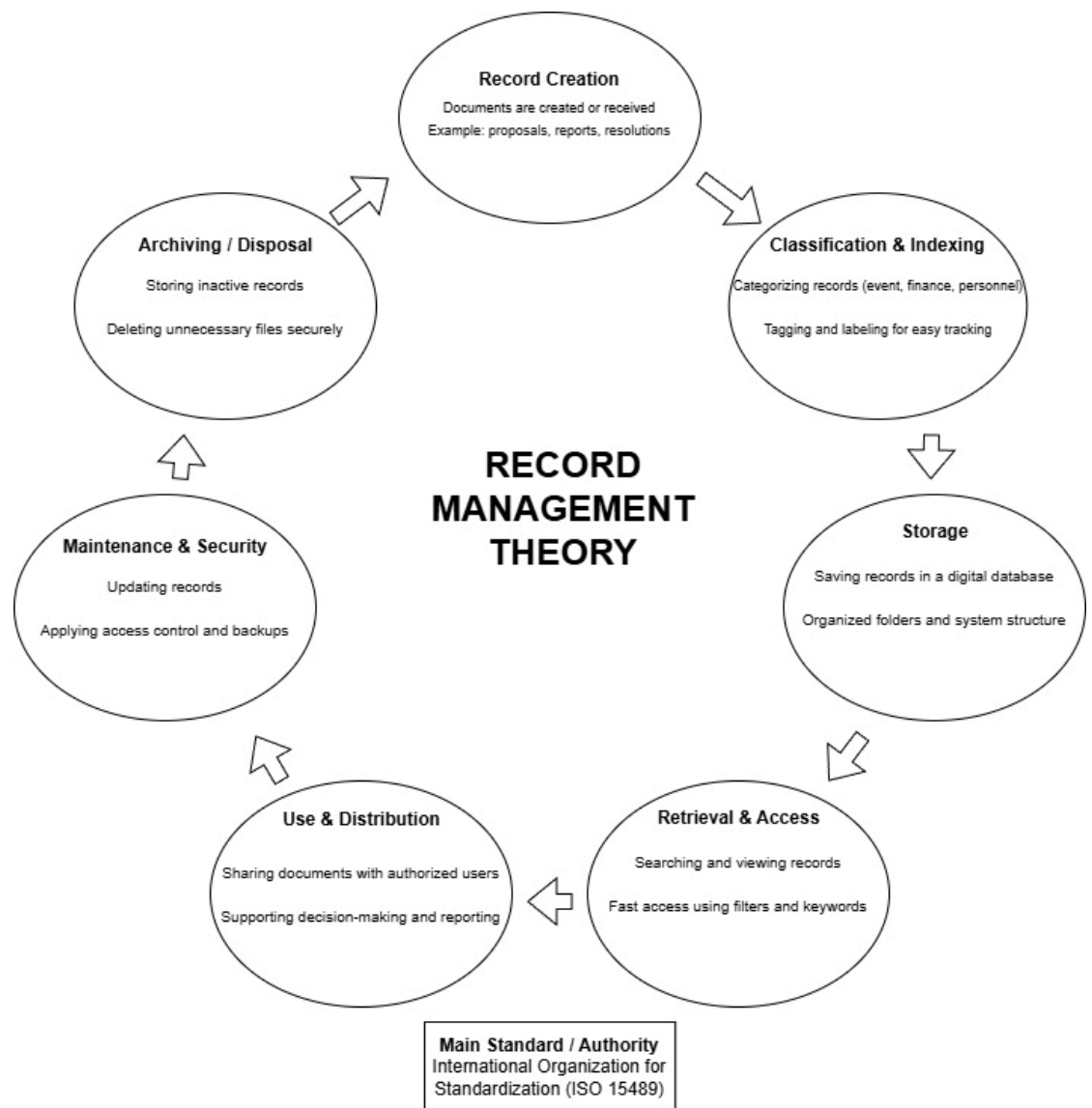


Figure 1: Record Management Theory

Figure 1 Illustrates the application of Records Management Theory in the proposed system through a continuous lifecycle process. It shows how records are created, classified, stored, retrieved, used, maintained, and eventually archived or disposed of in a structured and organized manner. Each stage ensures that records remain accurate, secure, and easily accessible throughout their lifecycle. The system supports efficient document handling by enabling quick retrieval, controlled access, and proper storage, while also maintaining data integrity through security and backup

mechanisms. This process allows the organization to effectively manage information, support decision-making, and ensure accountability and transparency in all operations.

Conceptual Framework

This framework demonstrates how the system transforms manual, paper-based operations into a centralized, automated, and secure digital environment. It emphasizes the flow of information from user-provided data and system interactions to processed outputs ensuring transparency, accountability, and operational efficiency within the Sangguniang Kabataan offices of selected barangays in Ormoc City.



Figure 2 : Conceptual Framework

Figure 2 presents the Input–Process–Output (IPO) model of the proposed SKActHub system. The input stage includes knowledge resources, hardware requirements, user credentials, personnel information, messages, and event details necessary for system operation. These inputs are processed through key system functions such as login and authentication, personnel updates, message handling, event management, notifications, analytics, and security with access control. Each process ensures that data is validated, organized, and protected. As a result, the system produces outputs in the form of a fully functional SKActHub platform that supports the monitoring and tracking of SK activities, promotes efficient governance, and enhances transparency and decision-making within the Ormoc City Youth Development Office.

Statement of the Problem

The Sangguniang Kabataan (SK) plays an important role in youth governance and in implementing programs within the community. To perform these responsibilities effectively, proper management of documents and member records is necessary to ensure transparency, accountability, and smooth operations. However, many SK offices in Ormoc City still rely on manual and paper-based systems. Because of this, documents are often lost, misplaced, or duplicated, and retrieving important files can take a lot of time. It is also difficult to monitor the status of submitted proposals and activities, especially when multiple steps and approvals are involved.

Given these challenges, this study seeks to answer the following questions:

- What kind of web-based system can be developed to better organize, store, archive, and manage SK documents and member information?

- What features should be included to make uploading, categorizing, searching, and retrieving documents easier and faster?
- How can the system help track the status of documents from submission to review and approval to improve workflow monitoring?
- How can a centralized account management feature help authorized users monitor the status of SK members?.
- What security measures, such as user authentication and role-based access control, are needed to protect sensitive documents and records?
- How effective is the proposed system in terms of improving the accessibility of data, organization and storage of documents, monitoring and tracking of activities, security and protection of records, and overall usability for SK officials?

Objectives of the Study

Project Goal

The main goal of this study is to design and develop SKActHub, a web-based Digital Document Management, Archiving, and SK Account Management System. The system aims to improve the efficiency, security, accessibility, and monitoring of official documents and SK member records for the Sangguniang Kabataan of Ormoc City. To achieve this goal, the study pursues the following objectives:

- To design and develop a web-based system that can organize, store, archive, and manage SK documents and member information efficiently

- To develop system features that enable users to upload, categorize, search, and retrieve documents efficiently, while ensuring that all documents undergo an approval process before final storage.
- To implement a document tracking system that monitors the status of documents from submission to review and approval.
- To develop a centralized SK account management module that enables authorized users to view all members and monitor their status.
- To incorporate security features such as user authentication, role-based access control, and data protection to ensure the safety of documents and records.
- To evaluate the effectiveness of the system in terms of usability, efficiency, accessibility, security, and accuracy in managing documents and member information.

Significance of the Study

For the Sangguniang Kabataan (SK) Officials of Ormoc City

The proposed system, SKActHub, is developed to enhance the efficiency, organization, and overall productivity of SK officials in Ormoc City. By transitioning from manual and paper-based processes to a centralized digital platform, the system enables more systematic storage, management, and retrieval of documents. This reduces the time and effort required to locate files and organize records, allowing officials to focus more on planning, implementing, and monitoring youth-related programs. Furthermore, the system supports improved workflow through real-time tracking of documents and activities, contributing to a more streamlined and effective governance process.

For the SK Chairperson, Secretary, and Treasurer

SKActHub provides reliable and well-structured information that supports informed decision-making among key SK officials. The Chairperson, Secretary, and Treasurer can efficiently review proposals, monitor accomplishment reports, and oversee the status of document approvals. Additionally, the system offers a centralized database of SK member records, enabling easy tracking of member status, whether active, inactive, or no longer in office. This ensures greater accuracy in record-keeping, minimizes errors, and promotes transparency and accountability within the organization.

For the Ormoc City Youth Development Office (OCYDO)

The Ormoc City Youth Development Office (OCYDO) will benefit from improved access to organized and digitized SK records through the implementation of SKActHub. The system enables faster and more convenient review of proposals, reports, and other documents submitted by various SK offices. As a result, the YDO can more effectively monitor youth programs, ensure compliance with established guidelines, and make data-driven decisions based on accurate and up-to-date information.

For Future Researchers

This study serves as a valuable reference for future researchers who aim to develop digital solutions for local government units and similar institutions. It offers insights into the design, development, and implementation of a web-based document management and monitoring system. Moreover, it provides a foundation for further innovation, as future researchers may enhance the system by integrating advanced

features, improving functionality, or expanding its application to other areas of digital governance and records management.

Scope and Delimitations of the Study

This study focuses on the design and development of SKActHub, a web-based Project Management, Document Management, and SK Account Management System intended to support the monitoring and tracking of Sangguniang Kabataan (SK) activities and SK membership information in order to strengthen SK governance for the Ormoc City Youth Development Office.

Scope

This study focuses on the design, development, and implementation of SKActHub for the Sangguniang Kabataan (SK) of Ormoc City. The system is intended to support the efficient management, archiving, organization, retrieval, and monitoring of documents related to SK activities. These documents include activity proposals, accomplishment reports, approvals, budgets, minutes of meetings, financial reports, resolutions, and other official SK records.

The implementation and testing of the system are limited to selected barangays in Ormoc City, specifically Barangay South, Barangay East, Barangay Ipil, Barangay Cagbuhangin, and Barangay Don Felipe Larrazabal. These barangays serve as the primary participants and pilot users of the system, allowing for proper evaluation and validation of its functionality and effectiveness.

In addition, the system includes a centralized SK account and membership management feature. This allows authorized users to view SK accounts, access a complete list of SK members, and monitor each member's status whether active,

inactive, or no longer part of the organization. This feature aims to improve accuracy in record-keeping and enhance administrative monitoring.

SKActHub implements a role-based access system intended for the SK Chairperson, SK Secretary, and SK Treasurer. Each user is provided with an individual account. The SK Chairperson has full access to all documents and system features, while the SK Secretary and SK Treasurer can manage and access only the documents they have personally submitted, based on their assigned responsibilities.

SKActHub mainly focuses on document posting, project and activity monitoring, account viewing, and membership tracking. It enables authorized users to upload, store, organize, retrieve, and monitor various documents related to SK activities while keeping member records updated. Through these features, the system aims to improve the monitoring of submitted documents, completed activities, and the participation of SK members.

Lastly, the system is developed as a web-based application using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js, based on the developers' technical knowledge and expertise. Since it is web-based, the system is designed to be accessed through a web browser and requires a desktop or laptop computer with a stable internet connection.

Delimitations

This study focuses on the development and implementation of SKActHub for the Sangguniang Kabataan (SK) of Ormoc City, specifically covering five selected barangays: Barangay South, Barangay East, Barangay Ipil, Barangay Cagbuhangin, and Barangay Don Felipe Larrazabal. It does not include all SK offices in Ormoc City, as

doing so would be too broad for a capstone project. Instead, the selected barangays provide a manageable scope that still reflects the actual needs and processes of SK offices, allowing the researchers to effectively develop and evaluate the system.

SKActHub is developed as a web-based system intended for use on desktop and laptop computers through a web browser. It is not optimized for mobile devices, which may limit accessibility for users who rely on smartphones. The system implements role-based access for key SK officials, namely the SK Chairperson, SK Secretary, and SK Treasurer, all of whom are provided with individual accounts.

The SK Chairperson is granted full access to all documents within the system, allowing comprehensive oversight and management. In contrast, the SK Secretary and SK Treasurer have limited access, where they can only view, manage, and retrieve the documents they have personally submitted or stored. This role-based structure ensures proper control, maintains data security, and clearly defines responsibilities among users while supporting efficient system management.

The system mainly focuses on document management, activity monitoring, and tracking the status of SK members. It does not include more advanced features such as automated approval workflows, financial tracking systems, attendance monitoring, role-based access control beyond the defined users, performance evaluation tools, offline functionality, real-time analytics, or integration with other government information systems. These limitations are intentional, as the system is designed to prioritize essential features that support efficient record-keeping and basic operational monitoring rather than complex, enterprise-level automation.

Definition of Terms

- ***Centralized SK Management Platform*** - A digital platform utilized by SK officials in selected barangays to upload, organize, monitor, and manage activity proposals, accomplishment reports, and member information efficiently.
- ***Participating SK Offices*** - The SK offices in selected barangays of Ormoc City that use SKActHub to manage documents and monitor youth membership records.
- ***System Administrator*** - The primary user of SKActHub in each barangay who uploads documents, monitors activities, manages membership status, and controls access permissions.
- ***Document and Records Assistant*** - A secondary user of SKActHub who supports the SK Chairman by uploading documents, updating member information, and maintaining activity records.
- ***Financial Records Handler*** - A secondary user of SKActHub who supports the SK Chairman by managing financial records, uploading financial documents, and maintaining budget and expense reports related to SK activities.
- ***Participation Classification Indicator*** - The designation used by SK officials to track member participation and eligibility in programs and activities.
- ***Planned Activity Record*** - A proposal prepared and submitted by the SK to document planned youth programs or events for monitoring and reference.
- ***Completed Activity Record*** - A report used by SK officials to document completed programs and evaluate activity performance.
- ***Document Handling Feature*** - The feature within SKActHub that enables SK officials to manage and access digital documents efficiently.

- ***Document Handling Feature*** - A system feature that allows the SK Chairman to assign and manage access privileges of the SK Secretary and SK Treasurer within SKActHub. The Chairman can monitor and access all uploaded documents, whereas the Secretary and Treasurer are limited to viewing and managing only the documents they submitted or the folders they created.
- ***Browser-Based System Interface*** - The interface of SKActHub accessed via desktop or laptop devices where users perform document management and monitoring tasks.
- ***MERN Stack*** - The technological foundation of SKActHub that enables data storage, system processing, and interactive dashboard functionality.
- ***Submission and Tracking Process*** - The process used by SK officials to ensure that proposals and reports are properly recorded, updated, and accessible.

CHAPTER II

Review of Related Literature

Effective document management and activity monitoring play a vital role in promoting transparency, accountability, and operational efficiency in institutions such as the Sangguniang Kabataan (SK). Proper record-keeping ensures that programs, financial transactions, and official activities are well-documented, providing reliable references that support informed decision-making and strengthen public trust. However, many SK offices still rely on traditional paper-based filing systems, which often result in misplaced documents, delayed retrieval, duplication, and limited accessibility. These manual processes also increase the risk of errors and physical damage to records, making it more difficult to monitor activities efficiently.

In many cases, SK officials spend unnecessary time searching for documents or verifying records, which slows down daily operations and affects productivity. This situation can also lead to inconsistencies in reporting and challenges in tracking the progress of activities. As a result, the overall management of records and monitoring of programs becomes less organized and more prone to inefficiencies.

Studies show that digital document management systems significantly improve record organization, security, and accessibility. Centralized platforms allow faster document retrieval, better classification, and more secure storage of files. Features such as automated tracking and role-based access control further enhance transparency by ensuring that only authorized personnel can access sensitive information. Despite these advantages, there are still limited studies focused on integrated systems specifically

designed for SK operations, highlighting the need for a platform that combines document management and activity monitoring to improve administrative efficiency.

Local Studies

Digital Document Management Systems

Dizon and Cruz (2023) developed a blockchain-based barangay document management system integrated with Optical Character Recognition (OCR) technology in a Philippine local government office. The study included 50 barangay staff members who utilized the system for six months. Their findings revealed that automation reduced manual encoding errors by 40%, accelerated document processing by 30%, and improved record immutability through blockchain technology, preventing unauthorized alterations. The study emphasizes that the integration of emerging technologies enhances public trust, transparency, and data integrity within local government operations, supporting the rationale for SKActHub's secure storage design.

Similarly, Innovatus Research Group (2019) implemented a web-based document management system in a barangay office in Leyte. Using pre- and post-implementation assessments, they found notable improvements in document classification, retrieval speed, and inter-office coordination. Document retrieval time was reduced from an average of 30 minutes under a paper-based system to under 5 minutes with the digital system. The study highlights that digital repositories reduce physical storage dependency and improve administrative efficiency, directly supporting SKActHub's document storage and organization functionality.

Reyes et al. (2021) studied cloud-based document archiving in municipal offices and reported enhanced compliance with record-keeping policies, reduced document

misplacement by 65%, and faster access to records. The study reinforces the importance of centralized document repositories, which SKActHub provides for SK activity proposals, accomplishment reports, and membership data.

Document Tracking and Workflow Monitoring

Bautista et al. (2022) developed a web-based document tracking system for a Philippine institutional setting involving 75 employees over six months. The system allowed real-time monitoring of documents from submission to approval, incorporating automated status updates, email and SMS notifications, and visual tracking dashboards. The study reported a 35% reduction in processing delays, improved coordination among departments, and faster completion of multi-step approval workflows. These findings support the inclusion of SKActHub's automated status updates and notification modules to improve the efficiency of SK document handling.

Similarly, Lopez and Mendoza (2023) implemented a barangay request and tracking system in Leyte, designed to minimize human errors, generate automated performance indicators, and enhance transparency. Their evaluation indicated that the system increased accountability as each document could be traced from submission to completion, reducing misplaced files and confusion during approval.

Santos et al. (2021) examined workflow monitoring in several Philippine municipal offices. Their research indicated that real-time dashboards and automated alerts facilitated faster decision-making, improved compliance with reporting timelines, and allowed administrators to detect bottlenecks in activities. These studies collectively validate SKActHub's inclusion of document tracking and workflow monitoring as essential components for operational transparency and administrative efficiency.

Cloud Computing in Document Management

Garcia and Lim (2020) explored the adoption of cloud-based document management systems in government offices and found that cloud technology significantly reduced infrastructure and maintenance costs. Their study showed that organizations no longer needed extensive physical storage or on-site servers, as documents could be securely stored in remote servers. This allowed employees to access files anytime and anywhere, improving flexibility and productivity. The system also enhanced collaboration, as multiple users could work on the same document without the need for physical file transfers.

Information Systems in Local Governance

Navarro and Perez (2022) examined the implementation of integrated information systems in Philippine local government units and found that these systems significantly improved coordination between departments and reduced delays in document processing. Their study demonstrated that centralized platforms enable data consolidation, allowing multiple offices to access the same information in real time. This reduced redundancy in records and minimized inconsistencies in reporting. Furthermore, the system improved workflow efficiency by eliminating repetitive manual encoding and simplifying data retrieval processes, which contributed to faster administrative operations.

Villanueva et al. (2023) further supported these findings by analyzing the use of information systems in municipal offices. Their study revealed that systems with centralized dashboards improved monitoring capabilities by allowing administrators to track activities, documents, and performance indicators in a single interface. This reduced the need for manual follow-ups and improved communication between

departments. The researchers concluded that integrated systems enhance operational efficiency and accountability, particularly in environments where multiple processes occur simultaneously.

Local Literature

Torres (2021) emphasized that information systems contribute to evidence-based governance by providing accurate, timely, and accessible data. The study highlighted that real-time information allows government officials to monitor ongoing activities, evaluate program effectiveness, and make informed decisions. In the context of youth governance, such as the Sangguniang Kabataan, the availability of reliable data ensures that projects and activities are properly documented and evaluated. The use of digital systems also improves transparency, as records can be easily reviewed and verified when needed.

Foreign Studies

Digital Document Management Systems

and Fianty (2023) investigated web-based document management systems in multiple Southeast Asian municipalities, involving 100 administrative users. Their study highlighted that centralized storage, workflow automation, and role-based access control significantly improved operational efficiency and interdepartmental collaboration. Audit trail features enabled supervisors to monitor document modifications, access history, and user activities, which increased transparency and minimized disputes between departments. The researchers also emphasized that digital systems reduced redundancies caused by duplicate records and improved reporting accuracy by providing real-time updates on the status of requests and documents. These

findings demonstrate that integrating workflow automation and role-based permissions can enhance accountability, a feature directly relevant to SKActHub's design.

Similarly, Smith et al. (2022) conducted a European study on local government digital archives, analyzing the effects of digital systems on record retrieval and data security. Their study found that digital archiving reduced document retrieval time by 60% and enhanced compliance with international data security standards, such as GDPR. The system also included version control, which prevented accidental or unauthorized modifications, ensuring document integrity. These findings highlight the importance of secure, centralized storage and automated access control in managing sensitive records.

Document Tracking and Workflow Monitoring

Awang Gani et al. (2024) investigated electronic document management systems in Malaysian government agencies. Their study highlighted that workflow monitoring strengthened compliance with institutional procedures, reduced the risk of lost documents, and improved audit readiness. The system's automated logging mechanisms provided verifiable trails for all activities, ensuring accountability and simplifying audits.

O'Connor and Lee (2022) studied digital workflow systems in Canadian municipal offices, observing that integrated dashboards and alerts facilitated task tracking, improved coordination across departments, and enabled administrators to detect and resolve process delays early.

Cloud Computing in Document Management

Chen et al. (2021) investigated cloud-based systems in government institutions and highlighted their role in disaster recovery and data protection. Their findings revealed that automatic backup features ensured that documents were preserved even in cases of hardware failure, natural disasters, or accidental deletion. This capability is particularly important for government offices that rely heavily on documentation, as it prevents permanent data loss and ensures continuity of operations.

Alotaibi and Federico (2021) examined the security aspects of cloud-based document management systems and found that features such as encryption, multi-factor authentication, and role-based access control significantly improved data protection. Their study reported a decrease in unauthorized access incidents and increased user confidence in the system. These security measures ensure that only authorized personnel can access sensitive information, which is critical in managing government records.

Rahman et al. (2023) also highlighted that cloud-based systems improve system scalability, allowing organizations to expand storage capacity without major infrastructure changes. Their study showed that scalable systems can accommodate increasing volumes of data, making them suitable for growing institutions. This flexibility ensures that document management systems remain efficient even as organizational needs evolve.

CHAPTER III

Methodology

Developing an effective technological solution requires a clear and well-organized approach. This chapter presents the methodology used in this study, including the processes involved in designing, developing, and evaluating SKActHub. It explains the research framework, system development approach, and the steps followed to ensure that the system addresses the challenges faced by the Sangguniang Kabataan in managing documents, monitoring activities, and tracking membership records. By following a structured method, the study aims to produce a system that is practical, reliable, and aligned with the needs of SK officials in the selected barangays of Ormoc City.

Project Design

This study used the Developmental Research Method, which focuses on the systematic design, development, implementation, and evaluation of a system to address specific organizational needs. Unlike descriptive research, which mainly describes existing conditions, developmental research aims to create and validate a technological solution that can improve current processes. This approach is suitable for information technology projects because it provides a clear and structured way of turning identified problems into a functional digital system.

In this study, the method guided the development of SKActHub by first identifying the challenges in manual document handling, activity monitoring, and membership record management within the Sangguniang Kabataan. These identified

issues served as the foundation for defining the system's objectives, features, and overall design.

The main goal of the study is to develop SKActHub, a web-based system for project management, document management, and SK account management for selected barangays in Ormoc City. The development process followed key stages such as planning, requirements analysis, system design, development, testing, and deployment to ensure that the system meets both user needs and operational requirements. The system is designed to provide secure document storage, organized record management, activity tracking, and centralized monitoring of SK members. Before implementation, the system was carefully tested and improved to ensure its functionality, usability, reliability, and security.

Development Model

The development of the system followed the Software Development Life Cycle (SDLC) to ensure a structured and organized process. In this study, the Agile Development Model was used because it allows flexibility, continuous feedback, and gradual improvement while the system is being developed.

The process started with gathering and analyzing requirements. During this stage, the researchers identified the needs of SK officials and reviewed how documents are currently managed in the selected barangays. This helped determine the important system features, types of documents to be handled, user roles, and the requirements for monitoring activities. Based on this information, the system design was created, including the overall structure of the system, database design, user interface, and workflow. Key features such as user accounts, document management, activity monitoring, and notifications were planned at this stage.

After the design phase, the system was developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. The development was done step by step, starting with core features like user authentication, document uploading, and activity management. Additional features such as notifications and dashboards were added afterward. This gradual approach made it easier to improve and adjust the system as development progressed.

Once development was completed, testing was conducted to ensure that the system worked properly and was easy to use. Functional testing and usability testing were performed to identify errors and areas that needed improvement. Any issues found were fixed, and changes were made based on feedback from potential users.

After testing, the system was deployed and implemented in selected barangays for actual use and evaluation. Finally, the system underwent continuous monitoring and improvement. The researchers observed its performance, addressed any issues encountered, and made further enhancements based on user feedback. This ongoing process helped ensure that the system remained effective, reliable, and responsive to the needs of its users.

Entity Relationship Diagram (ERD)

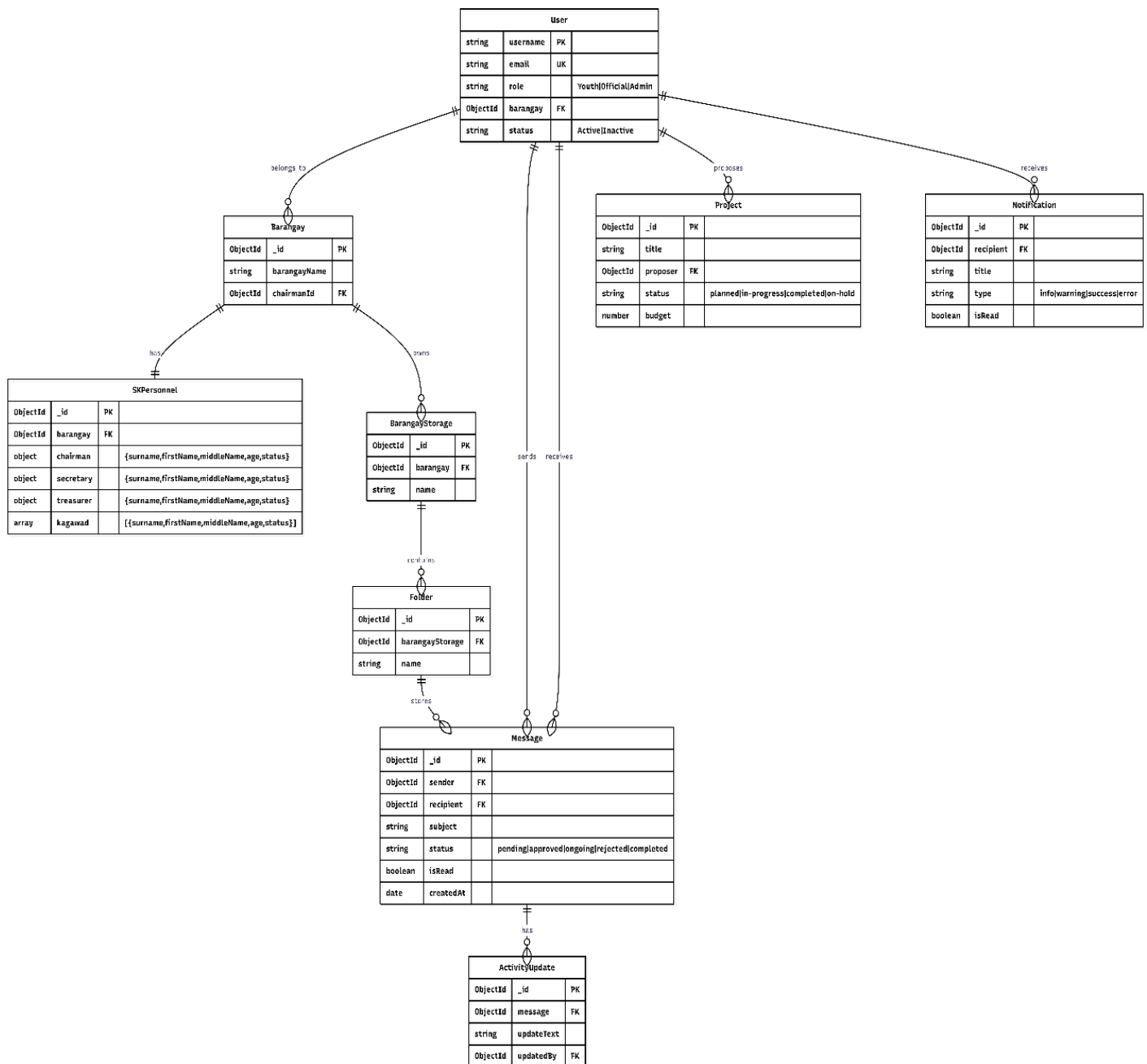


Figure 3: Entity Relationship Diagram

Figure 3 shows the overall structure of the SKActHub system database and how its main entities are connected. It highlights key components such as User, Barangay, SK Personnel, Projects, Messages, Notifications, and Storage, which work together to manage records, communication, and activities. The diagram illustrates how data flows

within the system, ensuring organized information handling and efficient operations for SK officials.

DATA FLOW DIAGRAM

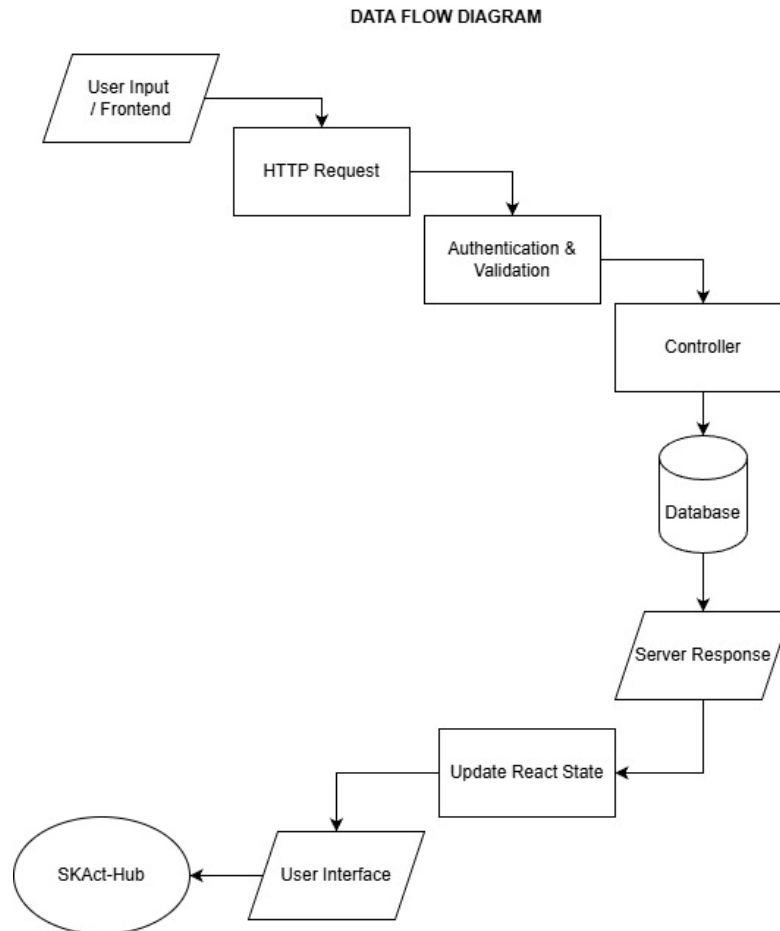


Figure 4: Data flow Diagram for SKAct-Hub

Figure 4 This figure illustrates the flow of data within the SKAct-Hub system. It begins with user input from the frontend, which is sent as an HTTP request to the server. The request then undergoes authentication and validation to ensure security and correctness of the data. Once validated, the request is processed by the controller, which interacts with the database to store or retrieve necessary information. After processing, a server response is generated and sent back to the frontend, where the React state is

updated accordingly. Finally, the updated data is reflected in the user interface, allowing users to view the results of their actions within the system.

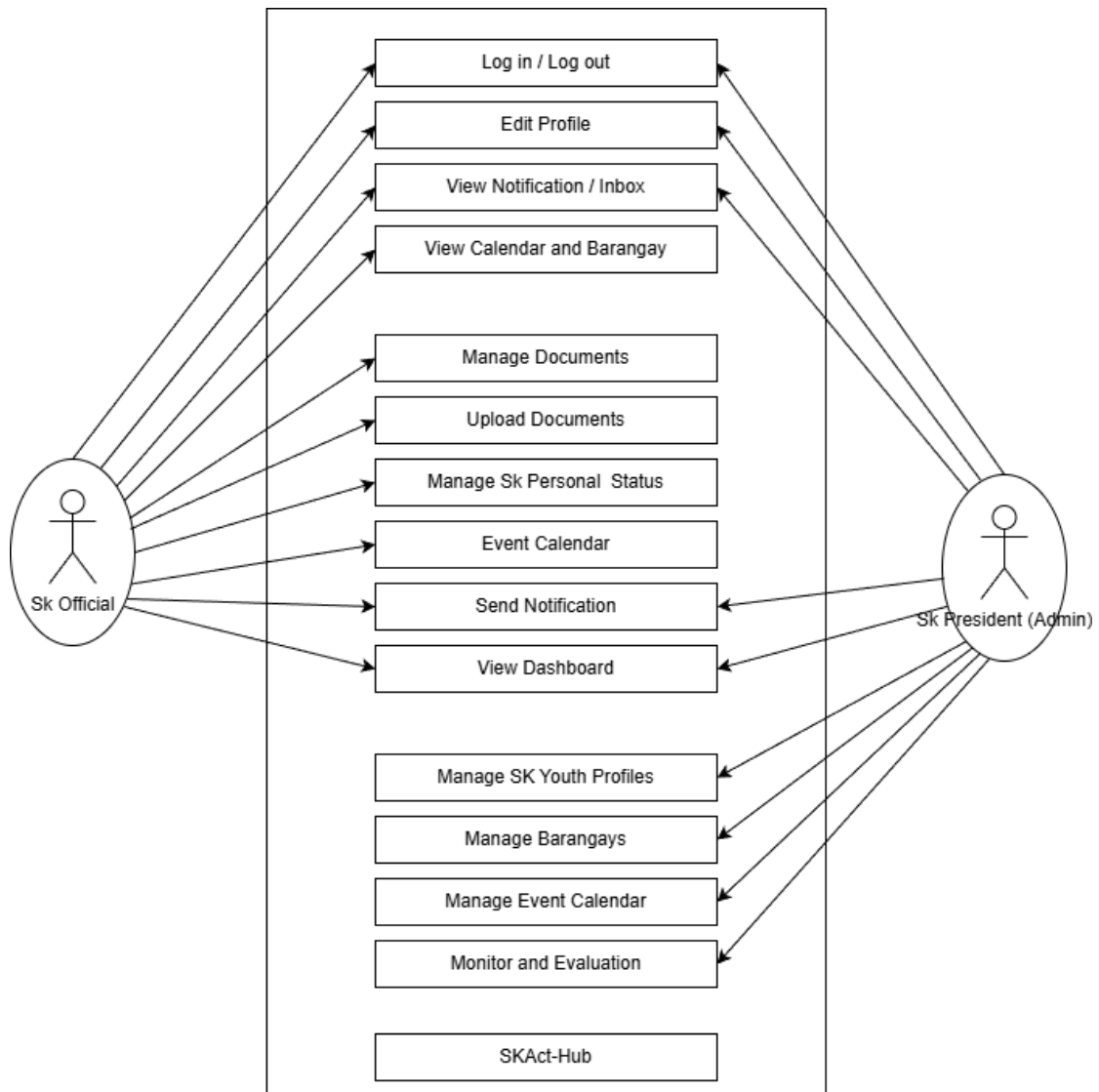


Figure 5: Use Case Diagram for SKAct-Hub

Figure 5 illustrates the interactions between the SK Official and SK President (Admin) with the SKAct-Hub system. It shows the main functions available in the system such as login/logout, managing documents, viewing notifications, event calendar, and dashboard. The diagram also highlights the additional administrative

functions of the SK President, including managing youth profiles, barangays, and monitoring activities.

System Architecture

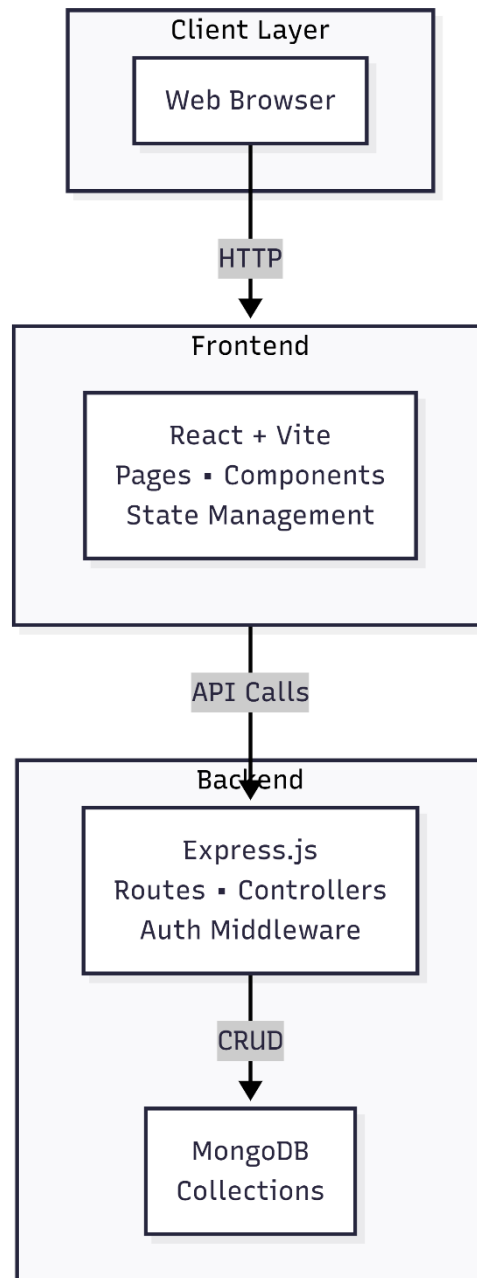


Figure 6: System Architecture

Figure 6 shows how the different parts of the system work together in a simple and connected flow. The process starts when the user opens the system in a web

browser, which then interacts with the frontend built using React and Vite to display pages and handle user actions. The frontend sends requests to the backend using API calls, where Express.js processes the data, applies security checks, and communicates with the MongoDB database to store and retrieve information.

A. Project Locale

This study was conducted in five (5) selected barangays in Ormoc City, Leyte, Philippines. Ormoc City is a component city located in the western part of Leyte Province in Eastern Visayas, known for being an important center for local governance, business, and community activities. The barangays included in this study are South, East, Ipil, Cagbuhangin, and Don Felipe Larrazabal. These areas were chosen because they are accessible to the researchers, have active Sangguniang Kabataan (SK) officials, and still rely mostly on manual or paper-based systems in managing their documents.

In these barangays, SK officials usually keep records such as activity reports, financial documents, meeting minutes, and member information in folders, notebooks, or filing cabinets. While this method has been used for a long time, it often leads to misplaced documents, difficulty in finding records, and delays in completing tasks. It can also make it harder for officials to track activities and stay organized, especially when handling multiple responsibilities.

Because of these challenges, this study aims to introduce the SKActHub system, which is designed to make document management easier and more efficient. By using this system, SK officials can store, organize, and monitor their records digitally, making information easier to access and manage. In the long run, this can help improve coordination among officials, speed up administrative work, and support better service for the youth in these communities.

B.Respondents of the Project

The respondents of the study are the official members of the Sangguniang Kabataan (SK) from five selected barangays. This includes five (5) SK Chairpersons, five (5) SK Secretaries, five (5) SK Treasurers, and (2) SK Kagawads, with each barangay represented by its respective set of officials. These participants were chosen because of their direct involvement in SK operations and document management within their communities.

The SK Chairperson serves as the leader of the organization and is responsible for overseeing youth programs and activities. Their role includes presiding over meetings, approving proposals and reports, and ensuring that all projects are properly implemented in accordance with existing rules and regulations. Meanwhile, the SK Secretary is primarily responsible for handling documentation and records, such as preparing meeting minutes, organizing proposals and accomplishment reports, and maintaining files to ensure that all documents are properly recorded and easily accessible. The SK Treasurer, on the other hand, manages the financial aspects of the SK, including handling budgets, recording financial transactions, and preparing financial reports to ensure proper use of funds.

In addition, the SK Kagawads play an important role in supporting the implementation of SK programs and activities. They assist in planning and executing projects, participate in decision-making, and help monitor ongoing activities within the barangay. Their involvement also includes contributing to the preparation and submission of necessary documents and reports.

These respondents were selected because they are directly involved in the preparation, submission, organization, and monitoring of SK-related documents and

activities. Their roles provide them with firsthand experience of the current processes used in their barangays. As a result, their feedback is essential in evaluating the effectiveness, usability, and overall functionality of the proposed SKActHub system.

C. Determination of the Sample Size

The study involved a total of 30 respondents selected from five barangays. The participants included SK Chairpersons, SK Secretaries, SK Treasurers, and SK Kagawads, with each barangay represented by its set of officials. These individuals were chosen because they are directly involved in SK operations and document management, making them the most suitable sources of information for the study.

The researchers used this group of respondents to support both quantitative and qualitative approaches. This allowed the collection of numerical data through structured responses, as well as deeper insights based on the participants' experiences and perspectives. By including key SK officials, the study ensured that the data gathered is relevant, reliable, and reflective of actual SK administrative practices.

D. Research Instrument

This study used both quantitative and qualitative methods to gather data from the respondents. For the quantitative part, a structured questionnaire with checklist and rating-scale questions was created and distributed through Google Forms. This made it easier for the respondents to answer and for the researchers to quickly collect and organize the data. The questions mainly focused on evaluating SKActHub in terms of usability, efficiency, functionality, and overall performance.

To support the numerical data, the questionnaire also included open-ended questions. These allowed the respondents to freely share their thoughts, experiences,

and suggestions while using the system. This gave the researchers a better understanding of what users felt about the system beyond just the ratings.

By combining both survey responses and written feedback, the study gathered both measurable results and real user insights. This helped provide a more complete and meaningful evaluation of SKActHub, especially in identifying its strengths and areas that still need improvement.

E. Project Description

SKActHub is a web-based system designed to enhance the management of Sangguniang Kabataan (SK) documents, activities, and membership information. The system enables SK officials to efficiently upload, organize, and archive important records such as activity proposals, accomplishment reports, and approval documents in a centralized digital repository. It also provides real-time monitoring of submitted documents and ongoing activities, allowing officials to track progress, identify delays, and ensure timely completion of tasks. In addition, SKActHub includes a membership management feature that allows administrators to maintain and update SK member records, including monitoring membership status such as active, inactive, or resigned members. To ensure data security and confidentiality, the system incorporates role-based access control, restricting access to sensitive information based on user roles and responsibilities. By transitioning from traditional paper-based processes to a structured digital workflow, SKActHub improves transparency, strengthens accountability, and enhances overall operational efficiency within SK offices.

Technology Used

The SKActHub system utilizes a combination of modern web technologies to ensure efficient performance, scalability, and user-friendly interaction. At the core of the system is MongoDB, a NoSQL document-oriented database that stores SK documents, member records, activity logs, and other system data in flexible JSON-like structures. Supporting the backend is Node.js, a runtime environment that executes server-side code, along with Express.js, a lightweight framework responsible for handling routing, API requests, and overall server-side logic. These technologies work together to process system operations and connect the database with the frontend interface.

On the client side, React.js is used to build dynamic and interactive user interfaces, enabling the creation of dashboards, document upload pages, and monitoring tools. To enhance the design and responsiveness of the system, Tailwind CSS is implemented as a utility-first UI framework, ensuring a clean and modern layout across devices. Communication between the frontend and backend is facilitated by Axios, which allows seamless HTTP requests and data exchange.

For data management and structure, Mongoose serves as an Object Data Modeling (ODM) library that simplifies database operations and enforces schema organization. Security is maintained through JSON Web Token (JWT), which provides a token-based authentication system to secure user access and manage sessions effectively. Additionally, Chart.js is used to present system data visually through charts and analytics, improving monitoring and decision-making.

To support development and collaboration, the project uses:

- Git and GitHub for version control, enabling developers to track changes and manage the source code efficiently
- Visual Studio Code as the primary development environment for coding, debugging, and maintaining the system

Overall, these technologies collectively ensure that the SKActHub system is robust, secure, and capable of delivering a responsive and efficient user experience.

Statistical Treatment

To ensure systematic and objective interpretation of the collected data, appropriate statistical tools were applied to both the pre-survey and post-survey questionnaire responses.

For the Pre-Survey Questionnaire, frequency distribution and percentage were used to summarize the respondents' answers. These statistical measures helped identify the prevailing document management practices, common challenges, retrieval efficiency, and readiness for digital system adoption among the selected Sangguniang Kabataan (SK) offices.

The formula used to compute the percentage is:

$$\text{Percentage} = \frac{f}{N} \times 100$$

where:

f = frequency of responses

N = total number of respondents

This method allowed the researchers to determine the proportion of respondents selecting each response option and to establish baseline data before system implementation.

For the Post-Survey Questionnaire, a five-point Likert Scale was utilized to measure respondents' level of agreement regarding the usability, effectiveness, security, and overall performance of SKActHub. The responses were analyzed using the *Weighted Mean* to determine the average rating for each statement and each evaluation category.

The formula for the Weighted Mean is:

$$\text{Weighted Mean} = \frac{\sum(f \times x)}{N}$$

where:

f = frequency of each response

x = numerical value assigned to each Likert scale option

N = total number of respondents

The computed weighted mean was interpreted using the following scale:

Mean Range Verbal Interpretation

Table 1 presents the mean ranges and their corresponding verbal interpretations used to analyze the respondents' level of agreement.

4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 – 3.40	Neutral

1.81 – 2.60

Disagree

1.00 – 1.80

Strongly Disagree

In addition to the weighted mean, *Standard Deviation (SD)* was used to measure the variability or dispersion of responses. Standard deviation determines how close or spread out the responses are from the computed mean. A low standard deviation indicates that the responses are closely clustered around the mean, suggesting consistent evaluations among respondents. A high standard deviation indicates greater variability in responses.

The formula used for Standard Deviation is:

$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{N}}$$

where:

x = individual response value

$\bar{x}$ = mean

N = total number of respondents

By computing both the weighted mean and standard deviation, the study ensures a more comprehensive analysis of the system evaluation results. The weighted mean determines the overall level of agreement, while the standard deviation provides insight into the consistency of respondents' perceptions. These statistical tools collectively support an objective and reliable assessment of SKActHub's effectiveness in improving document management, activity monitoring, and SK member administration.

Functional and Non-Functional Feature

The SKActHub system is designed to support the administrative and operational needs of the Sangguniang Kabataan by providing a set of well-defined functional and non-functional features. These features ensure that the system not only performs its intended tasks effectively but also maintains quality, security, and reliability in its overall performance.

The functional features focus on the core services and operations of the system. These are the specific actions that users can perform to manage SK activities and documents efficiently. In general, the system enables users to:

- Securely access the platform through user authentication using registered usernames and passwords
- Manage user accounts, including creating, updating, and assigning roles or permissions
- Upload important SK documents such as proposals, resolutions, and accomplishment reports
- Store documents in a centralized and secure digital repository for easy access
- Organize files into categories like proposals, reports, approvals, and financial records
- Search for documents quickly using filters such as title, date, or category
- Track the status of documents and activities (e.g., pending, approved, completed)
- Monitor ongoing SK activities and view real-time updates

- View and download documents as needed by authorized users
- Receive notifications regarding updates, approvals, or status changes
- Maintain user activity logs to ensure accountability and transparency
- Archive inactive documents securely, allowing retrieval when necessary while preventing accidental modification or deletion

On the other hand, the non-functional features describe the quality standards and system attributes that ensure a smooth and reliable user experience. These features do not focus on specific actions but rather on how well the system performs. The system is designed to:

- Provide fast performance, with pages and documents loading within a few seconds
- Ensure strong security through authentication, authorization, and protected data storage
- Maintain reliability by minimizing system downtime and ensuring consistent operation
- Offer high usability with a user-friendly interface that is easy for SK officials to navigate
- Support scalability, allowing the system to handle more users and data without slowing down
- Maintain compatibility across modern web browsers such as Chrome, Edge, and Firefox

- Allow easy maintenance, making future updates and improvements manageable
- Perform regular data backups to prevent loss of important documents

Overall, these functional and non-functional features work together to make the SKActHub system efficient, secure, and dependable for managing SK-related activities and records.

CHAPTER IV

PRESENTATION AND ANALYSIS OF DATA

This chapter presents the data gathered during the testing and evaluation of the developed system. The results are organized and presented using tables and descriptive analysis to clearly show the performance and effectiveness of the system. The data collected from respondents and system testing are analyzed to determine whether the objectives of the study were achieved. The findings in this chapter serve as the basis for interpreting the results and assessing the overall functionality and usability of the developed system

Level of Agreement

Table 3 presents the summary of the pre-survey results based on the responses of thirteen (13) participants out of the target twenty-five (30) respondents. The data were analyzed using frequency and percentage to determine the current practices in document management, monitoring, communication, and system readiness among SK officials. The results indicate that most respondents still rely on manual or partially digital processes, which affect efficiency, accessibility, and overall workflow.

Table 3. Pre-Survey Summary of Current Practices (N = 13)

Indicators	Most Common Response	Percentage
Document Storage	Paper-based with some digital	90%
Difficulty in Locating Documents	Sometimes	55%
Document Retrieval Time	5–15 minutes	52%
Lost or Misplaced Documents	A few times	68%

Indicators	Most Common Response	Percentage
Membership Record Management	Manual Logbook	55%

Table 3.1 Monitoring and Reporting (N = 13)

Indicators	Most Common Response	Percentage
Activity Tracking Method	Printed Reports	60%
Ease of Generating Reports	Neutral	48%
Centralized Monitoring System	Yes	65%
Delays in Submission	Sometimes	53%

Table 3.2 Communication and Accessibility (N = 13)

Indicators	Most Common Response	Percentage
Communication Method	Messenger/Chat Apps	68%
Centralized Messaging System	Yes	70%
Accessibility of Documents	Moderately Accessible	45%

Table 3.3 System Needs and Readiness (N = 13)

Indicators	Most Common Response	Percentage
Need for Digital System	Strongly Agree	75%
Willingness to Adopt System	Yes	80%

Table 3: Document Management

Indicator	%	f
Document Storage	90%	12
Difficulty in Locating Documents	55%	7
Document Retrieval Time	52%	7
Lost or Misplaced Documents	68%	9
Membership Record Management	55%	7
TOTAL		29

Table 3.1: Monitoring and Reporting

Indicator	%	f
Activity Tracking Method	60%	8
Ease of Generating Reports	48%	6
Centralized Monitoring System	65%	8
Delays in Submission	53%	7
TOTAL		29

Table 3.2: Communication and Accessibility

Indicator	%	f
Communication Method	68%	9

Indicator	%	f
Centralized Messaging System	70%	9
Accessibility of Documents	45%	6
TOTAL		24

Table 3.3: System Needs and Readiness

Indicator	%	f
Need for Digital System	75%	10
Willingness to Adopt System	80%	10
TOTAL		20

Table 3.4 Combined Summary of Pre-Survey Results (N = 13)

Category	No. of Indicators	Total f	Total Responses	Percentage
Document Management	5	42	65	65%
Monitoring and Reporting	4	29	52	56%
Communication and Accessibility	3	24	39	62%
System Needs and Readiness	2	20	26	77%
TOTAL	14	115	182	63%

The data in Table 3 and its subsections show that most SK offices still depend on manual and semi-digital systems, which results in challenges such as time-consuming document retrieval, occasional loss of records, and limited accessibility. In terms of monitoring and reporting, the lack of a centralized system contributes to delays and inefficiencies in tracking activities and generating reports. These findings are further supported by the overall results, which indicate a total weighted distribution of 14, 115, and 182, with an overall percentage of 63%. This suggests that while some level of digital practice exists, many processes are still not fully optimized or systematized, reinforcing the need for an integrated system like SKActHub to improve efficiency, accuracy, and accessibility in SK operations.

Level of Agreement

Table 2 presents the post-survey evaluation of the SKActHub system based on the responses of sixteen (16) participants out of the expected twenty-five (30), yielding a response rate of 64%. The data were analyzed using the weighted mean and interpreted using the given descriptive rating scale.

The results show that respondents generally have a positive evaluation of the system across its major features. *Overall System Effectiveness* obtained the highest weighted mean of 4.40 (Strongly Agree), indicating that the system is perceived as highly effective. In contrast, *Activity Monitoring and Tracking* received the lowest weighted mean of 3.20 (Neutral), suggesting that respondents have mixed opinions regarding this feature.

Meanwhile, *System Usability and User Experience* (4.35) and *Document Management and Organization* (4.34) are both interpreted as Strongly Agree, reflecting high user satisfaction. On the other hand, *SK Member Management* (4.05) and *Security*

and *Access Control* (4.10) are interpreted as Agree, indicating that these features are acceptable but may still require improvement.

Overall, the computed general weighted mean of 4.07 (Agree) shows that the SKActHub system is generally functional and acceptable. However, certain areas particularly activity monitoring may still need enhancement to further improve the overall user experience.

Table 2. Post-Survey Evaluation of SKActHub (N = 16)

Section A: System Usability and User Experience

Rating	Freq
5	7
4	7
3	1
2	1
1	0
WM = 4.35	Total = 16(Strongly Agree)

Section B: Document Management and Organization

Rating	Freq
5	7
4	7
3	1
2	1

Rating	Freq
1	0
WM = 4.34	Total = 16(Strongly Agree)

Section C: Activity Monitoring and Tracking

Rating	Freq
5	2
4	5
3	6
2	2
1	1
WM = 3.20	Total = 16 (Neutral)

Section D: SK Member Management

Rating	Freq
5	5
4	7
3	3
2	1
1	0
WM = 4.05	Total = 16 (Agree)

Section E: Security and Access Control

Rating	Freq
5	6
4	7
3	2
2	1
1	0
WM = 4.10	Total = 16(Agree)

Section F: Overall System Effectiveness

Rating	Freq
5	8
4	6
3	2
2	0
1	0
WM = 4.40	Total = 16(Strongly Agree)

Indicators	WM	DR
1. System Usability and User Experience	4.35	Strongly Agree
2. Document Management and Organization	4.34	Strongly Agree
3. Activity Monitoring and Tracking	3.20	Neutral
4. SK Member Management	4.05	Agree
5. Security and Access Control	4.10	Agree
6. Overall System Effectiveness	4.40	Strongly Agree
Total Weighted Mean	24.44	
General Weighted Mean	4.07	Agree

Testing Results:

Table 4

Login Test

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC001	Login with valid credentials	1. Open login page 2. Enter valid email & password 3. Click login	User should be logged in and redirected to dashboard	User redirected successfully	☒ Pass
TC002	Login with invalid password	1. Enter correct email 2. Enter wrong password 3. Click login	Error message should appear	Error displayed: "Invalid password"	☒ Pass
TC003	Empty input fields	1. Leave email & password empty 2. Click login	Validation error should show	"Fields required" shown	☒ Pass

Figure 7 : Login Test

Table 5**Add Event**

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-E01	Login with valid admin credentials	System is running	1. Open login page 2. Enter username & password 3. Click Sign In	User redirected to Events page	Pass
TC-E02	Open Create Event modal	User is logged in	1. Click "Create Event" button	Modal should appear	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-E03	Empty form submission validation	Modal is open	1. Click Create Event without input	Browser validation message for required fields (e.g., title)	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-E04	Input event title and description	Modal is open	1. Enter title 2. Enter description	Inputs should accept values	Pass
TC-E05	Input start and end date	Modal is open	1. Enter start date 2. Enter end date	Dates should be accepted	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-E06	Select "Specific" visibility	Modal is open	1. Click "Specific" radio button	Barangay dropdown should appear	Pass

TC-E07	Barangay dropdown loads options	Dropdown visible	1. Wait for options 2. Check options count	Options should be loaded	Pass
TC-E08	Select barangay	Dropdown loaded	1. Select barangay option	Barangay should be selected	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-E09	Submit valid event form	All fields filled	1. Fill form correctly 2. Click Create Event	Success or response message should appear	Pass
TC-E10	Display submission feedback	Form submitted	1. Observe message area	System should show success or error message	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-E11	Close modal using Cancel button	Modal is open	1. Click Cancel	Modal should close	Pass

Figure 8 : Add Event Test

Edit Official

Table 6

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-EO01	Login with valid admin credentials	System is running	1. Open login page 2. Enter username & password 3. Click Sign In	User redirected to SK Officials page	Pass

TC-EO02	Open Edit Official modal	User is logged in	1. Click "More" button2. Click "Edit"	Edit modal should appear	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-EO03	First name required validation	Modal is open	1. Clear first name2. Click Save Changes	Browser validation message should appear	Pass
TC-EO04	Last name required validation	Modal is open	1. Clear last name2. Click Save Changes	Validation message should appear	Pass
TC-EO05	Position required validation	Modal is open	1. Clear/select empty position2. Submit form	Validation message should appear	Pass
TC-EO06	Barangay required validation	Modal is open	1. Clear/select empty barangay2. Submit form	Validation message should appear	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-EO07	Update basic fields	Modal is open	1. Enter new first name2. Enter new last name3. Select position4. Select barangay	Fields should update successfully	Pass

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-EO08	Open password reset fields	Modal is open	1. Click "Reset Password"	Password fields should appear	Pass
TC-EO09	Password mismatch validation	Password fields visible	1. Enter password 2. Enter different confirm password 3. Submit	Error "Passwords do not match" should appear	Pass
TC-EO10	Correct password confirmation	Password mismatch fixed	1. Match confirm password with password	Error should disappear	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-EO12	Cancel button closes modal	Modal is open	1. Click Cancel	Modal should close	Pass

Figure 9 : Edit Official Test

Create User Official

Table 7

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-O01	Login with valid admin credentials	System is running	1. Open login page 2. Enter username & password 3. Click Sign In	User redirected to SK Officials page	Pass

TC-O02	Open Create SK Official modal	User is logged in	1. Click "Add Official" button	Modal should appear	Pass
TC-O03	Empty form validation	Modal is open	1. Click Create without input	Validation errors should appear (e.g., First name required)	Pass
TC-O04	Input first and last name	Modal is open	1. Enter first name2. Enter last name	Inputs should accept values	Pass
TC-O05	Username auto-generation	Names entered	1. Enter first & last name2. Check username field	Username should be auto-generated	Pass
TC-O06	Dropdown selection (Position & Barangay)	Modal is open	1. Select position2. Select barangay	Dropdown values should be selectable	Pass
TC-O07	Password mismatch validation	Modal is filled	1. Enter password2. Enter different confirm password3. Submit	Error message "Passwords do not match" should appear	Pass
TC-O08	Correct password confirmation	Password exists	1. Match confirm password with password	Error should disappear	Pass
TC-O09	Password strength indicator	Password field filled	1. Enter password2. Observe UI	Password strength indicator should be visible	Pass
TC-O10	Toggle password visibility	Password field filled	1. Click toggle button	Password visibility should change	Pass
TC-O11	Submit valid form	All fields valid	1. Fill all fields correctly2. Click Create Official	Modal should close (success)	Pass
TC-O12	Prevent submission if errors exist	Form has validation errors	1. Try submitting invalid form	Form should not submit	Pass

TC-O13	API error handling (No token)	User logged out (token removed)	1. Remove token 2. Refresh page	System should show error / unauthorized behavior	Pass
TC-O14	API failure feedback	Submission fails	1. Submit form 2. Simulate API error	Error message/banner should appear	Pass

Figure 10 : Create User Official

Folder Creation

Table 8

Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-F01	Cancel button closes Create Folder modal	User is logged in and inside Barangay Storage	1. Click "New Folder" 2. Click "Cancel"	Modal should close	Pass
TC-F02	X button closes Create Folder modal	Same as above	1. Open Create Folder modal 2. Click "X" button	Modal should close	Pass
TC-F03	Create button disabled when input is empty	Modal is open	1. Do not input folder name 2. Check Create button	Button should be disabled	Pass
TC-F04	Create folder successfully	Modal is open	1. Enter folder name 2. Click Create	Folder should appear in list	Pass

TC-F05	Create folder using Enter key	Modal is open	1. Enter folder name 2. Press Enter	Folder should be created	Pass
TC-F06	Duplicate folder name is rejected	Folder already exists	1. Enter existing folder name 2. Click Create	Error message should appear ("already exists")	Pass
TC-F07	Open folder view modal	Folder exists	1. Click folder card	Folder modal should open with search bar	Pass
TC-F08	Search inside folder (empty state)	Folder view modal is open	1. Type random text in search 2. Observe result	"No documents found" message appears	Pass
TC-F09	Close folder view modal using Close button	Folder modal is open	1. Click "Close"	Modal should close	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-F11	Add Document button opens compose modal	Folder exists	1. Click "+" button	Compose modal should open	Pass
TC-F12	Compose submit disabled when fields empty	Compose modal open	1. Leave fields empty 2. Check button	Button should be disabled	Pass

TC-F13	Cancel button closes compose modal	Compose modal open	1. Click Cancel	Modal should close	Pass
TC-F14	X button closes compose modal	Compose modal open	1. Click X	Modal should close	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-F15	Set folder status to Ongoing	Folder exists	1. Click "Set Ongoing"2. Confirm	Status should change to Ongoing	Pass
TC-F16	Cancel status change	Status modal open	1. Click "Set Completed"2. Click Cancel	Status should remain unchanged	Pass
TC-F17	Set folder status to Completed	Folder exists	1. Click "Set Completed"2. Confirm	Status should change to Completed	Pass
Test Case ID	Test Scenario	Preconditions	Test Steps	Expected Result	Status
TC-F18	Cancel delete folder	Folder exists	1. Click Delete2. Click Cancel	Folder should NOT be deleted	Pass
TC-F19	Confirm delete folder	Folder exists	1. Click Delete2. Click Confirm	Folder should be removed from list	Pass
TC-F20	Cleanup test folder	Test folder exists	1. Delete test folder	Folder should be removed	Pass

User Acceptance Testing (UAT)

Before fully implementing the system, User Acceptance Testing (UAT) was carried out to make sure that it works well for its intended users. This step allowed actual SK officials to try the system and see if it fits their needs and daily tasks.

During this phase, selected users interacted with the system in a real-world setting. They tested its features, explored its functions, and shared their experiences while using it. This helped the researchers understand how easy the system is to use, how well it performs, and whether it meets user expectations.

The main goal of UAT is to confirm that the system is ready for use. It also provides an opportunity to gather feedback, identify minor issues, and make necessary improvements. Through this process, the researchers were able to ensure that the system is not only functional, but also practical, reliable, and user-friendly for SK officials.

A. Admin Module – SKActHub

Table 9

Test Scenario	Objective	Process	Expected Result	Actual Result	Status
1. Admin Login	To verify admin access	Enter credentials and click login	Redirect to dashboard	Successfully logged in	Passed
2. View Pending Messages	To view submitted documents	Open pending messages	Display message details	Displayed correctly	Passed
3. Approve Documents	To approve submissions	Click approve	Document stored properly	Successfully approved	Passed
4. Reject Documents	To reject invalid documents	Click reject	Document not stored	Successfully rejected	Passed
5. Create SK Official	To create accounts	Fill form and submit	Account created	Successfully created	Passed
6. Create Event Schedule	To create system events	Input event details and submit	Event saved	Event created successfully	Passed
7. User Logs Monitoring	To track system activity	Filter logs by name/type/date	Logs displayed	Logs shown correctly	Passed

B. SK Official Module – SKActHub

Table 10

Test Scenario	Objective	Process	Expected Result	Actual Result	Status
1. Official Login	To verify official access	Enter credentials	Redirect to dashboard	Logged in successfully	Passed
2. Inbox (Receive & Sent)**	To view communication records	Open inbox → check received & sent tabs	Messages displayed correctly	Messages shown properly	Passed
3. Create Event	To allow official to create events	Input title, description, start/end date & visibility	Event saved and displayed	Event created successfully	Passed
4. Barangay Folder Management	To manage document folders	Create or delete folder	Folder created/deleted	Successfully managed folders	Passed
5. Submit Document for Approval	To send documents to admin	Enter subject, details, attach file, then submit	Document sent to admin for approval	Document successfully submitted	Passed
6. SK Personnel Management	To manage SK member status	Input surname, first name, middle initial, age, set status (active/inactive)	Member record saved and updated	Member status updated successfully	Passed
7. View & Update Profile	To edit official profile	Update personal details and save	Changes reflected	Successfully updated	Passed
8. Logout	To securely exit system	Click logout	Redirect to login	Logged out successfully	Passed
9. Navigation & Usability	To evaluate ease of use	Navigate modules	Smooth navigation	User-friendly interface	Passed

CHAPTER V

SUMMARY FINDINGS, CONCLUSION AND RECOMMENDATIONS

Summary

This study focused on the development of SKActHub, a web-based system designed to improve document management, activity monitoring, and record organization within the Sangguniang Kabataan (SK). The researchers identified that many SK offices still rely on manual processes, which often result in misplaced documents, delays in approval, and difficulty in tracking records. To better understand these challenges, data were gathered through interviews, surveys, and observation involving SK officials such as the Chairman, Secretary, Treasurer, and other SK Kagawad.

Based on the findings, the researchers developed SKActHub as a centralized system with features such as document uploading with approval workflow, role-based access control, event management, and notifications. The system was built using the MERN stack and evaluated through User Acceptance Testing, where users found it functional, reliable, and easy to use. Overall, the system proved to be an effective solution that improves efficiency, enhances transparency, and supports better decision-making within the SK.

Findings

Based on the data gathered and the evaluation of the system, several important findings were identified. SK officials currently experience difficulties in managing documents due to their reliance on manual and paper-based processes. As a result,

documents are often misplaced, duplicated, or hard to retrieve, which affects overall efficiency. In addition, monitoring activities and tracking records take a significant amount of time and can sometimes be inconsistent, making it harder for officials to stay organized.

With the implementation of SKActHub, these challenges were significantly reduced. The system improved document organization by providing centralized digital storage, making it easier to manage and access files. The approval-based workflow added an extra layer of control by ensuring that only verified documents are stored, while role-based access control enhanced security by limiting user access based on their responsibilities. The system also performed well in terms of speed, especially in document retrieval and page loading, although document uploading takes slightly longer due to the approval process. Overall, users found the system easy to use, reliable, and helpful in completing their tasks more efficiently.

Conclusions

Based on the findings of the study, it can be concluded that the SKActHub system effectively addresses the common problems experienced in traditional document management and activity monitoring within the SK. The shift from manual, paper-based processes to a digital system provided a more organized, efficient, and secure way of handling records. Tasks that were previously time-consuming, such as searching for documents and tracking activities, became faster and more manageable through the system.

The use of a centralized digital platform significantly reduced the risk of lost, duplicated, or misplaced documents, while also making information more accessible to authorized users. The approval-based workflow played an important role in ensuring

that all documents are properly reviewed before being stored, which helps maintain accuracy and accountability. At the same time, the implementation of role-based access control strengthened system security by ensuring that users can only access data relevant to their responsibilities.

Overall, SKActHub proved to be a practical and reliable solution that improves daily operations within the SK. It not only enhances efficiency and organization but also promotes transparency and better decision-making among officials. The positive feedback from users during testing shows that the system is easy to use and meets their needs, making it suitable for real-world implementation and future adoption in other barangays.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are proposed:

For the Sangguniang Kabataan (SK) Officials: It is recommended that they fully adopt and consistently utilize the SKActHub system in their daily operations to enhance efficiency in document management, activity monitoring, and record organization.

For the System Users (SK Chairperson, Secretary, and Treasurer): Proper training and orientation programs should be conducted to ensure that users are equipped with the necessary knowledge and skills to effectively operate the system.

For Future System Developers: It is recommended to further enhance the system by incorporating additional features such as mobile compatibility, automated approval workflows, and integration with other government information systems.

For System Administrators: Regular system maintenance and updates should be implemented to ensure continuous functionality, data security, and overall system reliability.

For Stakeholders and Users: Continuous collection and evaluation of user feedback are encouraged to support ongoing system improvement and ensure that the platform remains responsive to user needs.

For Future Researchers: It is recommended to expand the scope of the study by implementing the system in other barangays or local government units and by exploring the integration of more advanced technologies to further improve digital governance.

Appendices & References:

References:

[1] Dizon, J. R., & Cruz, A. M.: *Blockchain-Based Barangay Document Management System with OCR Data Extraction*, IC-ITECHS Journal, 2023.
<https://jurnal.ubhinus.ac.id/index.php/IC-ITECHS/article/view/1712>

[2] Reyes, M. A., & Dela Cruz, J. P.: *Digitalizing Governance: A Transformation on the Processes of One Community in the Philippines*, International Journal of Multidisciplinary Research and Analysis (IJMRA), 2020.
<https://ijmra.in/v3i7/Doc/8.pdf>

[3] Santos, R. L., Mendoza, K. P., & Villanueva, J. T.: *Contextualized Online Document Management System*, Journal of Innovative Technology Convergence (JITC), 2021.
<https://innocn.innotcs.org/index.php/jitc/article/view/55>

[4] Bautista, K. L., Ramos, J. D., & Ortega, S. M.: *Web-Based Document Tracking System for Catanduanes State University*, Journal of Information Systems Engineering and Management (JISEM), 2022.
<https://www.jisem-journal.com/article/web-based-document-tracking-system>

[5] Triyana, A., & Fianty, M.: *Web-Based Document Management Systems in Educational Institutions*, International Journal of Science, Technology, and Management (IJSTM), 2023.
<https://ijstm.inarah.co.id/index.php/ijstm/article/view/1106>

[6] Bizi, F., Ghazali, N., & Yunus, M.: *Adoption of Electronic Document Management Systems in Organizations*, Journal of Information and Knowledge

Management(JIKM),2025.

<https://www.worldscientific.com/doi/10.1142/S021964922550003X>

[7] Awang Gani, R., Abd Kadir, M., Ab Rahman, S., & Mohd Yunus, N.: *Electronic Document Management in an E-Government Environment*, European Proceedings of Social and Behavioural Sciences, 2024.
<https://www.europeanproceedings.com/article/10.15405/epsbs.2024.01.02>

[8] Anonymous: *Factors Influencing Digital Adoption of Electronic Document Management Systems in the Public Sector*, ScienceDirect, 2025.
<https://www.sciencedirect.com/science/article/pii/S0740624X25000104>

[9] Lopez, A. M., & Mendoza, R. T.: *EaseDocument System: A Web-Based Platform for Barangay Requesting Document*, ResearchGate, 2023.
<https://www.researchgate.net/search?q=EaseDocument%20Barangay>

[10] Garcia, P. R., Navarro, C. J., & Lim, S. A.: *Docu-Go: The Development and Assessment of a Web-Based Barangay Document Requesting System*, ResearchGate, 2022.
<https://www.researchgate.net/search?q=Docu-Go%20Barangay>

[11] Innovatus Research Group: *Implementing Document Management System Technology in Barangay Paligui, Apalit, Pampanga*, Innovatus Research Publications, 2019.
<https://www.researchgate.net/search?q=Barangay%20Document%20Management%20Paligui>

[12] De Vera, R. J.: *E-RECORD: Web-Based Management System for Barangay Parian*, Studocu, 2021.

<https://www.studocu.com/ph/search?query=E-RECORD%20Barangay%20Parian>

[13] Braude, Eric J.: *Software Engineering: An Object-Oriented Perspective*, 5th Edition, McGraw-Hill, 2001, ISBN 0-07-365578-3.

<https://www.mheducation.com/highered/product/software-engineering-braude/M9780073655780.html>

[14] Venners, Bill.: *The Demand for Software Quality*, Artima Software Inc., August 12, 2005.

<http://www.artima.com/intv/serious.html>

[15] *Record Management Theory*, ScienceDirect.

<https://www.sciencedirect.com/topics/social-sciences/record-management>

[16] United Nations: *E-Government Survey 2022: The Future of Digital Government*, United Nations Department of Economic and Social Affairs, 2022.

<https://publicadministration.un.org/egovkb/en-us/reports/un-e-government-survey-2022>

[17] World Bank: *Digital Government Overview*, World Bank Group, 2021.

<https://www.worldbank.org/en/topic/digitaldevelopment/brief/digital-government>

[18] IBM Corporation: *Cloud Computing: A Complete Guide to Cloud Technology*, IBM, 2023.

<https://www.ibm.com/topics/cloud-computing>

[19] Microsoft Corporation: *What is Cloud Computing?*, Microsoft Azure Documentation, 2023.

<https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-cloud-computing/>

[20] Asian Development Bank: *4 Ways Governments Can Benefit from Cloud Computing*, Asian Development Bank, 2021.
<https://seads.adb.org/articles/4-ways-governments-can-benefit-cloud-computing>

[21] National Privacy Commission: *Data Privacy Act of 2012 (Republic Act No. 10173)*, Government of the Philippines, 2012.
<https://www.privacy.gov.ph/data-privacy-act/>

[22] Nielsen, J.: *Usability Engineering*. Morgan Kaufmann, 1993.
<https://www.nngroup.com/books/usability-engineering/>
(Supports usability evaluation in your system testing)

[23] Brooke, J.: *SUS: A Quick and Dirty Usability Scale*. In: *Usability Evaluation in Industry*, 1996.
<https://www.usability.gov/how-to-and-tools/methods/system-usability-scale.html>
(If you used survey evaluation—very strong citation)

[24] Pressman, R. S., & Maxim, B. R.: *Software Engineering: A Practitioner's Approach*, 9th ed., McGraw-Hill, 2020.
(Supports system development methodology / SDLC)

[25] Sommerville, I.: *Software Engineering*, 10th ed., Pearson, 2016. (Supports system design, architecture, and testing)

[26] ISO/IEC 25010:2011: *Systems and Software Quality Requirements and Evaluation (SQuaRE)*.

<https://iso25000.com/index.php/en/iso-25000-standards/iso-25010>

(Very important for system quality: usability, security, efficiency)

[27] Fielding, R. T.: *Architectural Styles and the Design of Network-Based Software Architectures*, Doctoral Dissertation, 2000.

<https://www.ics.uci.edu/~fielding/pubs/dissertation/top.htm>

(Supports REST API / MERN architecture)

[28] OWASP Foundation: *OWASP Top 10 Web Application Security Risks*, 2021.

<https://owasp.org/www-project-top-ten/>

(Strong support for your **Security and Access Control** feature)

[29] MongoDB Inc.: *What is MongoDB?*, 2023.
<https://www.mongodb.com/what-is-mongodb> (Supports your database choice in MERN stack)

[30] Meta (Facebook): *React Documentation*, 2023.
<https://react.dev/>

(Supports frontend framework used in your system)

[31] OpenJS Foundation: *Node.js Documentation*, 2023.
<https://nodejs.org/en/docs>

(Supports backend/server-side technology)

[32] Express.js: *Web Framework for Node.js*, 2023.
<https://expressjs.com/>

(Supports API/backend implementation)

Appendices A:

System Features:

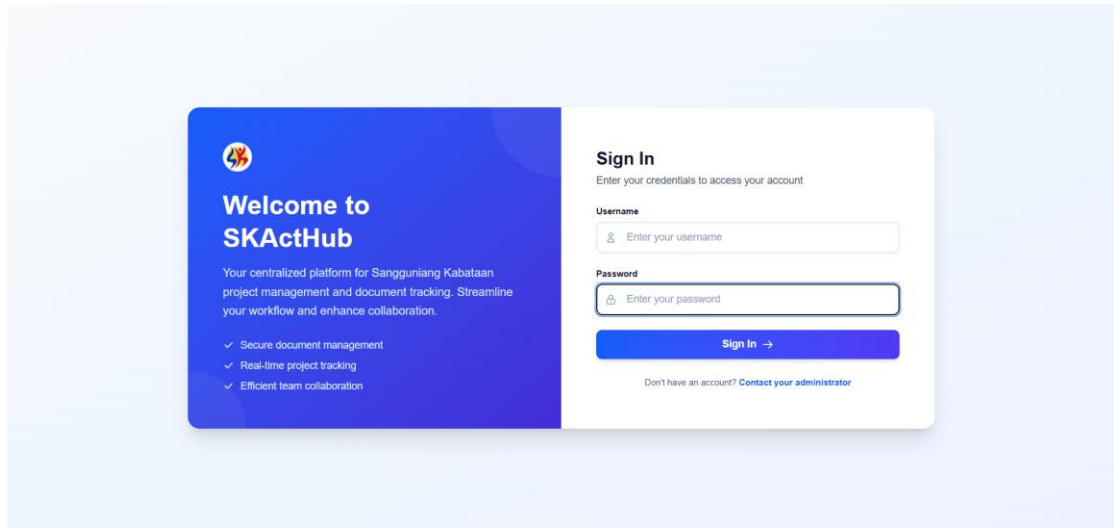


Figure 8 : Login Of SkAct-Hub

Admin Panel:

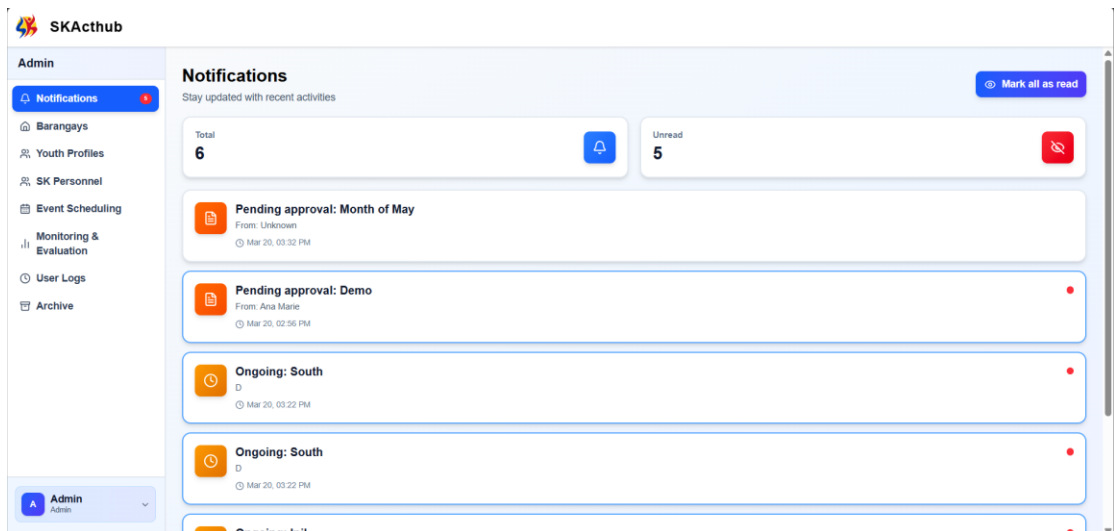


Figure 9 : Notification Of SkAct-Hub

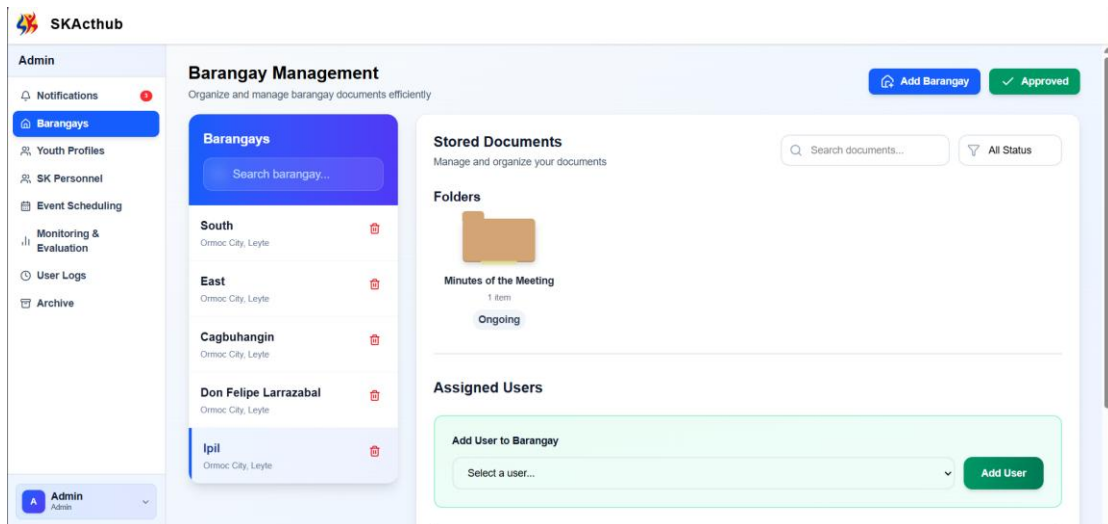


Figure 10 : Barangay Management Of SkAct-Hub

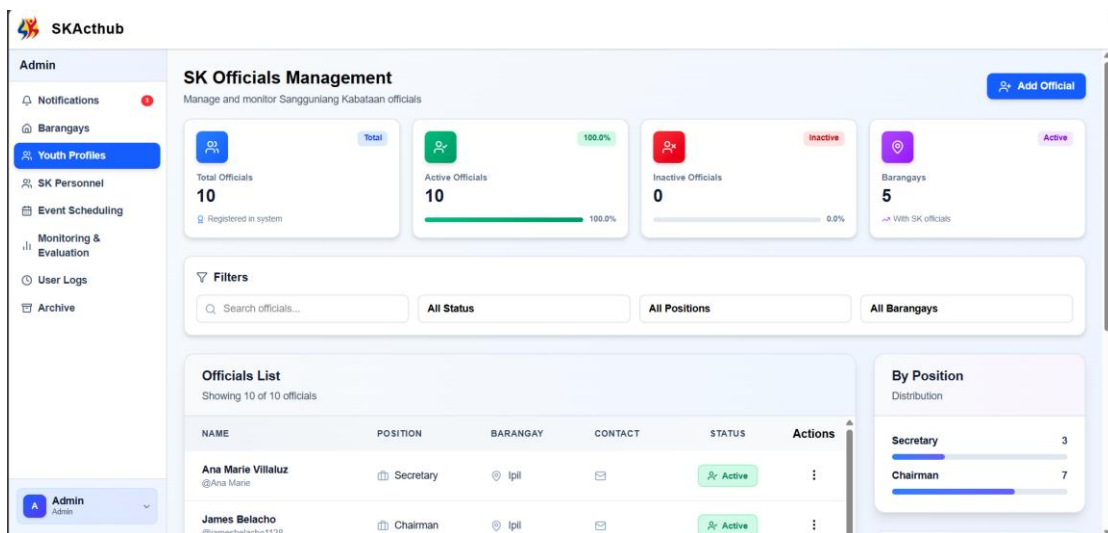


Figure 11 : User Management Of SkAct-Hub

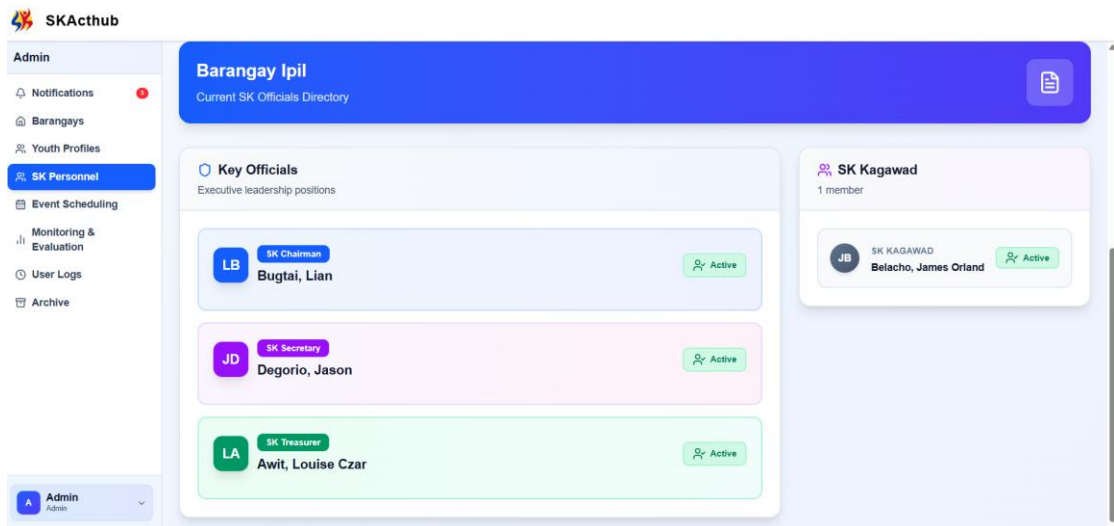


Figure 12: Sk Official Status Of SkAct-Hub

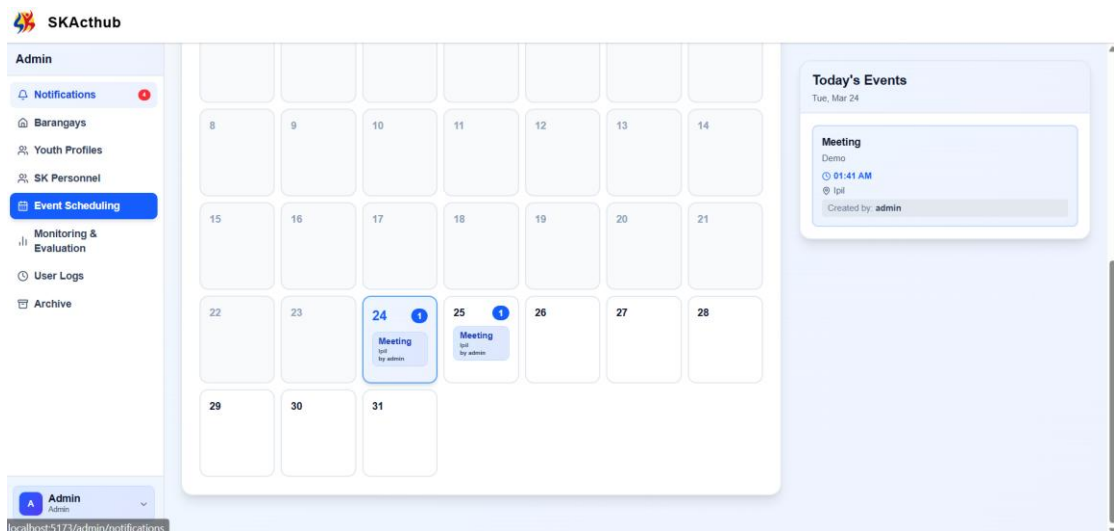


Figure 13 : Event Scheduling Calendar Of SkAct-Hub

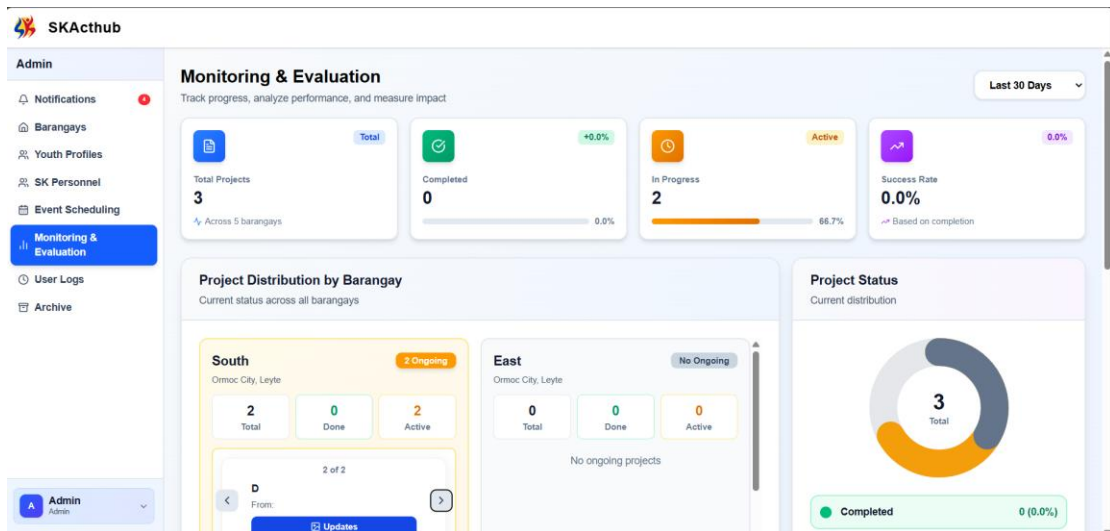


Figure 14 : Monitoring & Evaluation Of SkAct-Hub

User Logs
Track login, logout, and system actions across all barangays

Key Metrics:

- TOTAL LOGS: 579
- ACTION TYPES: 13
- TOP USER: @admin (Login 317, Edit User 91, Logout 80)

Search user or description... All actions dd/mm/yyyy to dd/mm/yyyy Export CSV

TIMESTAMP	USER	ACTION	DESCRIPTION	IP
Mar 24, 2026, 10:45 AM	Admin @admin	Login	Admin logged in	1
Mar 24, 2026, 10:45 AM	Admin @admin	Logout	Admin logged out	1
Mar 24, 2026, 10:15 AM	Ana Marie Villaluz @Ana Marie	Delete Message	Deleted message with subject: test	1
Mar 24, 2026, 10:15 AM	Admin @admin	Delete Message	Deleted message with subject: Demo	1
Mar 24, 2026, 10:15 AM	Admin @admin	Delete Message	Deleted message with subject: Meeting	1
Mar 24, 2026, 10:15 AM	Admin @admin	Delete Message	Deleted message with subject: Demo	1

Figure 15: User Logs Of SkAct-Hub

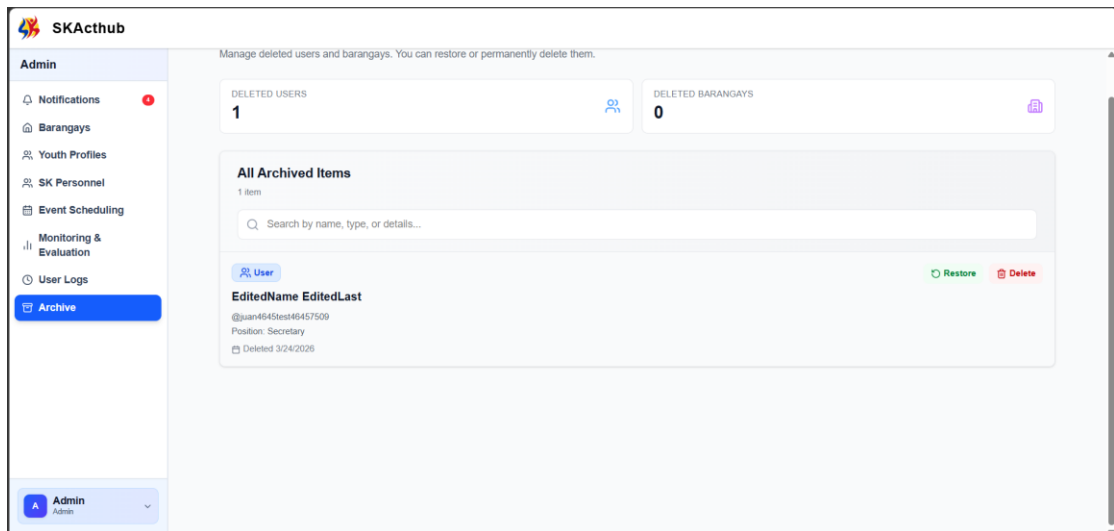


Figure 16 : Archive Page Of SkAct-Hub

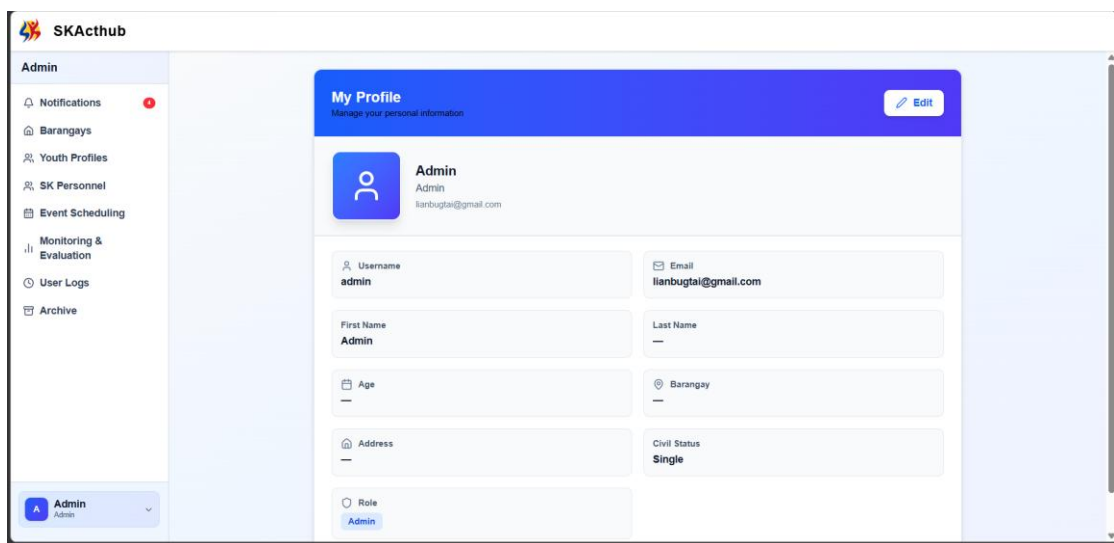


Figure 17 : View Profile Of SkAct-Hub

Official Panel :

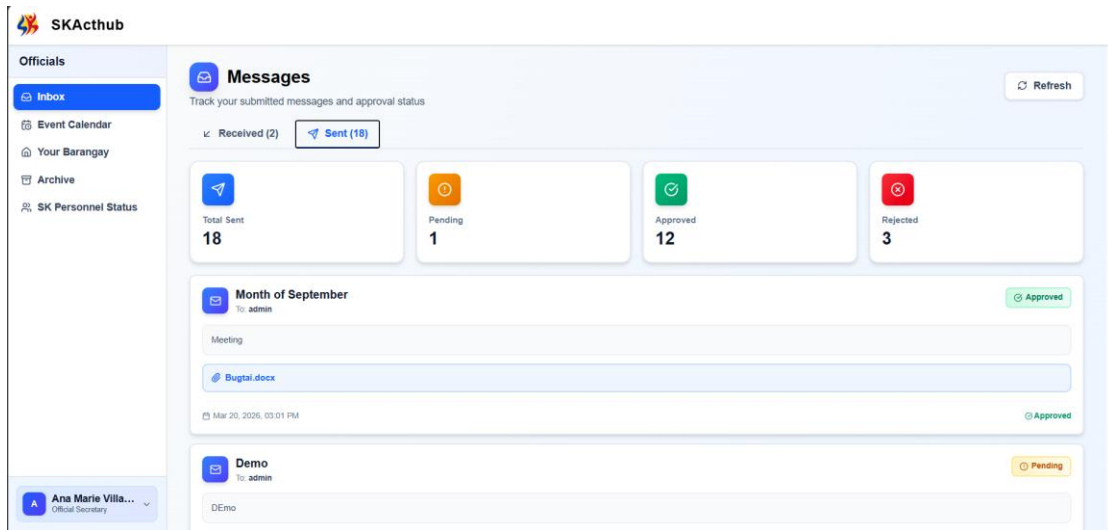


Figure 18 : Inbox of the Official user in SkAct-Hub

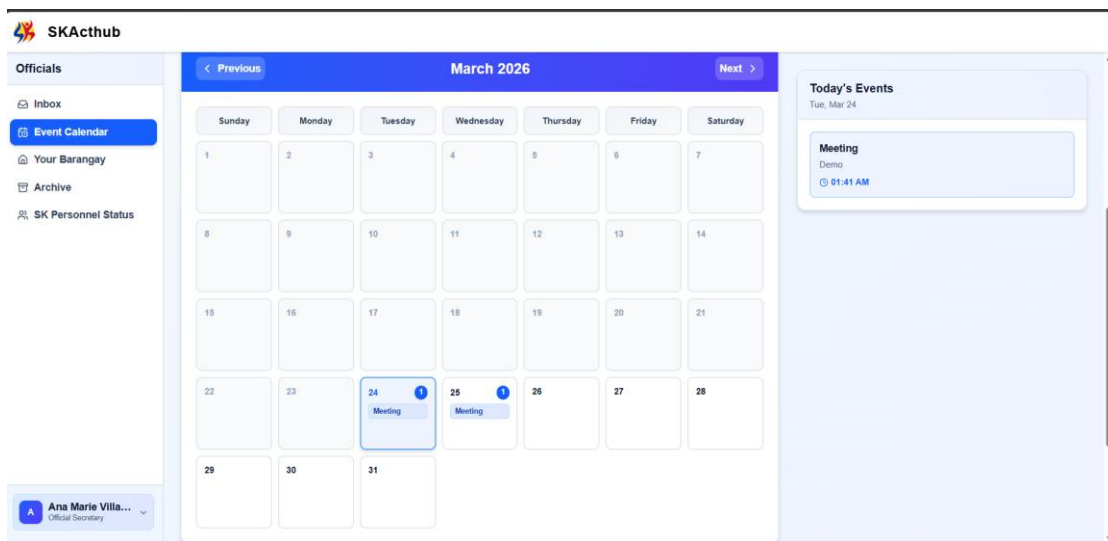


Figure 19 : Event Scheduling of the Official User in SkAct-Hub

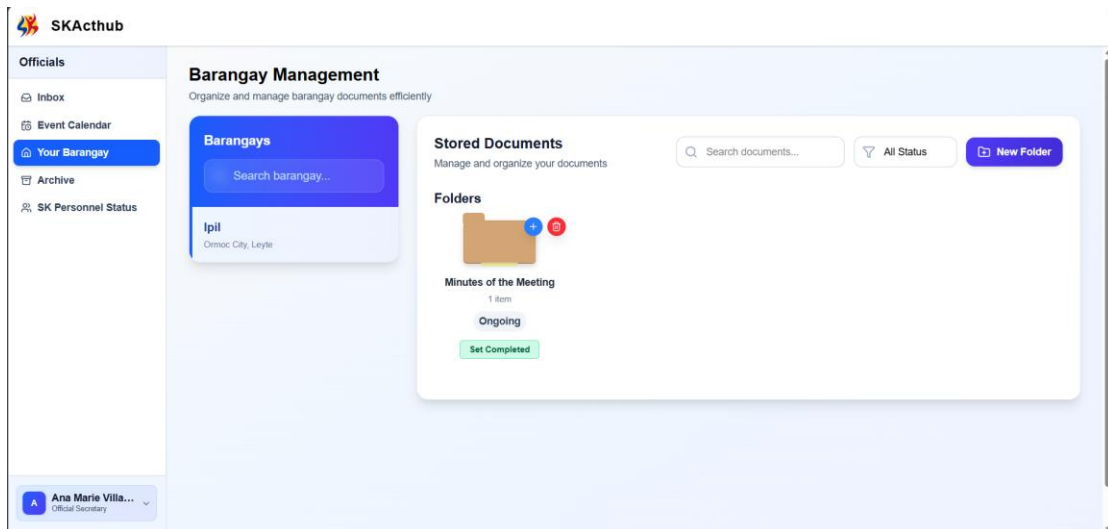


Figure 20 : Stored Document of the Official User in SkAct-Hub

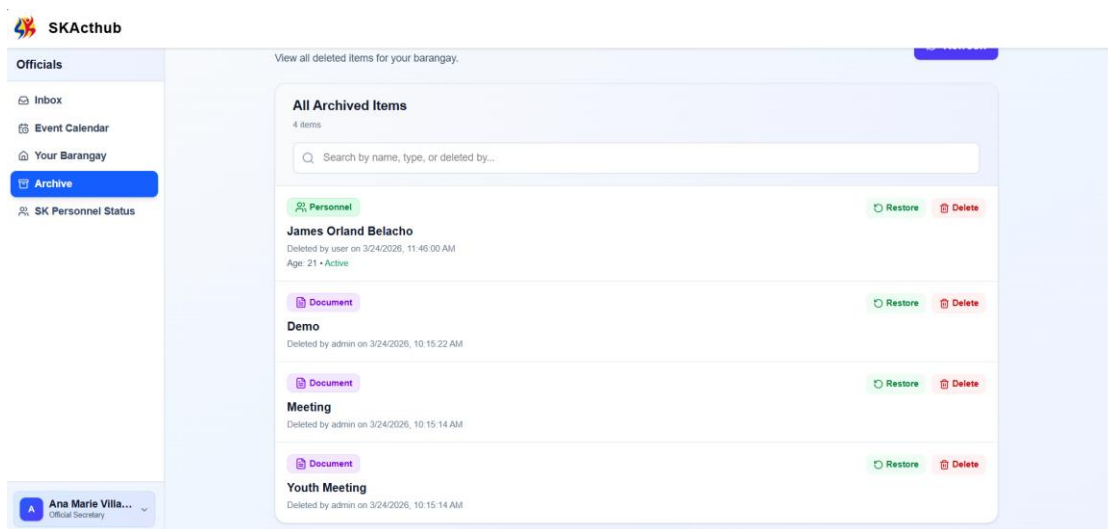


Figure 19 : Archive Page of the Official User in SkAct-Hub

POST SURVEY QUESTIONNAIRES

Name of Respondent: _____

Barangay: _____

Title: SKActHub: A Project Management for Monitoring and Tracking SK Activities to Strengthen SK Governance for the Ormoc City Youth Development Office

Instructions:

Please rate the following statements based on your experience using **SKActHub**.

Scale:

- 1 – Strongly Disagree
- 2 – Disagree
- 3 – Neutral
- 4 – Agree
- 5 – Strongly Agree

Please check (✓) one answer for each statement.

Section A: System Usability and User Experience

1. I can easily log in and access SKActHub.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

2. The system interface is clear and organized.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

3. I can easily navigate between different system modules.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

4. Uploading and categorizing new documents is easy.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

5. Searching for specific documents is simple and efficient.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

6. Overall, I am satisfied with the usability of SKActHub.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

Section B: Document Management and Organization

7. SKActHub effectively organizes SK documents (proposals, reports, approvals).

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

8. Document retrieval is faster than the previous paper-based system.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

9. The system reliably prevents document loss or misplacement.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

10. I am satisfied with the document storage and archiving process in SKActHub.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

11. The system allows me to manage multiple types of documents simultaneously.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

12. I rarely encounter errors when uploading or retrieving documents.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

Section C: Activity Monitoring and Workflow Tracking

13. I can easily track the status of submitted SK activities or proposals.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

14. Notifications and alerts about activity updates are accurate and timely.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree

- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

15. SKActHub makes monitoring ongoing SK projects simple and efficient.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

16. The system effectively summarizes completed and pending tasks.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

17. Activity tracking dashboards are clear and informative for decision-making.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

Section D: SK Member Management

18. Viewing and updating SK member information is easy.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

19. I can effectively track membership status (active, inactive, resigned).

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

20. SKActHub keeps SK member records accurate and up-to-date.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

Section E: Security and Access Control

21. My documents and SK member records are secure in SKActHub.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

22. User authentication and role-based access control function properly.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

23. I have not observed any unauthorized access or potential security issues.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

Section F: Overall System Effectiveness

24. SKActHub has improved document management, activity monitoring, and SK member management.

- ☐ 1 – Strongly Disagree
- ☐ 2 – Disagree
- ☐ 3 – Neutral
- ☐ 4 – Agree
- ☐ 5 – Strongly Agree

25. Please provide any suggestions or comments to further improve SKActHub:

PRE-SURVEY QUESTIONNAIRE

Title: SKActHub: A Project Management for Monitoring and Tracking SK Activities to Strengthen SK Governance for the Ormoc City Youth Development Office

Purpose: This survey aims to assess the existing document management practices, challenges, and system needs of the Sangguniang Kabataan (SK) offices prior to the implementation of SKActHub.

Part I: Respondent Profile

1. Name (Optional): _____
2. Position:
 - ☐ SK Chairperson
 - ☐ SK Kagawad
 - ☐ SK Secretary
 - ☐ SK Treasurer
 - ☐ SK Staff
3. Barangay: _____
4. Years in Service:
 - ☐ Less than 1 year
 - ☐ 1–2 years
 - ☐ 3–5 years
 - ☐ More than 5 years

Part II: Current Document Management Practices

Instruction: Please select the answer that best describes your current situation.

1. How are SK documents currently stored?
 - ☐ Purely paper-based
 - ☐ Paper-based with some digital copies
 - ☐ Mostly digital
 - ☐ Fully digital
2. How often do you experience difficulty locating important documents?
 - ☐ Never
 - ☐ Rarely
 - ☐ Sometimes

- ☐ Often
 - ☐ Always
3. How long does it usually take to retrieve a specific document?
 - ☐ Less than 5 minutes
 - ☐ 5–15 minutes
 - ☐ 16–30 minutes
 - ☐ More than 30 minutes
 4. Have you experienced lost or misplaced documents?
 - ☐ Never
 - ☐ Once
 - ☐ A few times
 - ☐ Frequently
 5. How are membership records managed?
 - ☐ Manual logbook
 - ☐ Excel or spreadsheet
 - ☐ Online system
 - ☐ No organized system

Part III: Monitoring and Reporting

6. How do you track SK activities and events?
 - ☐ Notebook/logbook
 - ☐ Printed reports
 - ☐ Spreadsheet
 - ☐ No formal tracking system
7. How easy is it to generate reports for submission to local authorities?
 - ☐ Very Easy
 - ☐ Easy
 - ☐ Neutral
 - ☐ Difficult
 - ☐ Very Difficult
8. Do you have a centralized system for monitoring project status?
 - ☐ Yes
 - ☐ No
9. How often do delays occur in document submission or reporting?
 - ☐ Never
 - ☐ Rarely
 - ☐ Sometimes

- ☐ Often
- ☐ Always

Part IV: Communication and Accessibility

10. How do SK officials usually communicate regarding activities?
- ☐ Face-to-face
 - ☐ Messenger/Chat apps
 - ☐ SMS
 - ☐ Email
 - ☐ Written memos
11. Is there a centralized messaging system for official communication?
- ☐ Yes
 - ☐ No
12. How accessible are important SK documents when needed?
- ☐ Very Accessible
 - ☐ Accessible
 - ☐ Moderately Accessible
 - ☐ Hard to Access
 - ☐ Not Accessible

Part V: System Needs and Readiness

13. Do you believe a digital document management system is necessary?
- ☐ Strongly Agree
 - ☐ Agree
 - ☐ Neutral
 - ☐ Disagree
 - ☐ Strongly Disagree
14. Which features would be most beneficial? (Select all that apply)
- ☐ Event management
 - ☐ Member record management
 - ☐ Messaging system
 - ☐ Document storage
 - ☐ Notifications
 - ☐ Analytics dashboard
15. Are you willing to adopt a centralized digital system for SK operations?
- ☐ Yes
 - ☐ No
 - ☐ Maybe

SKActHub System Performance Evaluation Survey

Section A: Respondent Information

1. Name (Optional)

2. Barangay / Office

Section B: System Evaluation

Instruction:

Please rate each statement based on your experience using SKActHub.

Please encircle the rate you like.

Scale:

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

System Speed

1. Document Uploading

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

2. Document Retrieval

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

3. Page Loading

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

System Security

4. User Login Security

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

5. Access Control

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

6. Document Approval

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

7. Data Protection

5 – Strongly Agree

4 – Agree

3 – Neutral

2 – Disagree

1 – Strongly Disagree

Section C: Feedback

8. Suggestions / Comments



CONTACT

☎ +639513641101

✉ lianbugtai@gmail.com

📍 Bryg. Ipil, Osco Compound

EDUCATION

2022- Present

ACLC COLLEGE OF ORMOC

- Bachelor of Science in Information Technology

2020 - 2022

SAINT'S PETERS COLLEGE OF ORMOC

- Humanities and Social Sciences (HUMSS)

2016 - 2020

SAINT'S PETERS COLLEGE OF ORMOC

2010 - 2016

IPIL CENTRAL SCHOOL

LIAN EMMANUEL V. BUGTAI

PROFILE

A motivated IT student with a strong foundation in computer systems, front-end programming, and networking principles. Demonstrated ability to collaborate in team projects and manage tasks using leadership and time management skills. Familiar with HTML, CSS, and basic troubleshooting. Eager to gain hands-on experience through an IT internship and contribute to real-world projects.

SKILLS

- Fast- Learner
- Teamwork
- Good Communication
- Critical Thinking
- Front-End Programming
- Microsoft Office Suite (Word, Excel, PowerPoint)
- Basic Python

PROJECTS

Freelance Marketplace System (School Project)

Tools used: M.E.R.N. Stack

Collaborated with a team to develop a web-based platform that connects freelancers with job posters. The system allows users to create accounts, search for job listings, apply for jobs, and manage ongoing work through a user-friendly dashboard.

- Implemented user authentication for both freelancers and employers.
- Designed job search and filtering features to enhance usability.
- Built job posting, application, and status-tracking functionalities.
- Focused on responsive UI and intuitive navigation to ensure accessibility across devices.

REFERENCE

Engr. Ma. Lelena Trinidad

OJT Adviser

ACLC College of Ormoc City, Inc.

Phone: +639703789767

Ellen M. Guinares

IT Instructor

ACLC College of Ormoc City, Inc.

Phone: +639619713200

LOUISE CZAR I. AWIT

 Talisay, Matag-ob, Leyte

 09935872560 / 09561293276

 awitlouise16@gmail.com



ABOUT ME

I am an enthusiastic and goal-driven Bachelor of Science in Information Technology student with a genuine interest in programming, system development, and solving technical challenges. With a foundation in software development, networking concepts, and database handling, I am excited to immerse myself in hands-on experiences. Recognized for being a quick learner, and flexible, I am dedicated to continuous improvement and eager to make meaningful contributions to a forward-thinking company during my on-the-job training.

PROFILE

Age: 22 years old
Birthdate: 05-12-2003
Sex: Male
Civil Status: Single

PERSONAL SKILLS

- Strong communication and interpersonal skills
- Adaptable and quick learner
- Problem-solving mindset
- Strong Attention to Detail
- Goal-Oriented
- Time management and organizational skills

TECHNICAL SKILLS

- Computer Literate
- Basic knowledge in programming languages
- Familiar with web development fundamentals
- Troubleshooting and basic computer hardware/software maintenance
- Understanding of database concepts

EDUCATION

- ACLC COLLEGE OF ORMOC
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
2022-PRESENT
- LA CONSOLACION COLLEGE-LILOAN
SCIENCE TECHNOLOGY ENGINEERING MATHEMATICS
SENIOR HIGH SCHOOL
2020-2022
- LA CONSOLACION COLLEGE-LILOAN
JUNIOR HIGH SCHOOL
2016-2020

LANGUAGE

- English
- Tagalog
- Bisaya

REFERENCES

ENGR. MA. LELENA TRINIDAD
OJT Adviser
ACLC College of Ormoc City, Inc.
Ormoc City
09703789767

ELLEN M. GUINARES
IT Instructor
ACLC COLLEGE OF ORMOC CITY, INC.
09619713200



Contact

Phone

0962-602-6020

Email

rollslawie@gmail.com

Address

Brgy. Mabato, Ormoc City

Education

2018 - 2020 (Junior High)
Ormoc Immaculate Conception School
Foundation Inc.

2021 - 2022 Senior High
San Lorenzo Ruiz College of Ormoc Inc.

2022 - Present
ACLC College of Ormoc Inc.

Skills

- Graphic Designing
- Video Editing
- Documenting

Computer Skills

- Text Processor
- Slide Presentation

Tools Used

- Microsoft Word
- Capcut
- Canva

Language

English

JOHNLOE G. ARCILLAS

ACLC COLLEGE OF ORMOC

with [X]+ years of experience in developing and executing data-driven marketing strategies that increase brand visibility, customer engagement, and revenue. Skilled in digital marketing, content creation, campaign management, and market research. Proven track record of delivering ROI-focused solutions and collaborating across cross-functional teams to drive business growth.

Experience

2022 - 2023

ACLC College of Ormoc

Department Secretary

- Accurately record, transcribe, and distribute meeting minutes for departmental and faculty meetings, maintaining a clear and organized record of discussions, decisions, and action items.
- Serve as the primary administrative support for the department head, ensuring smooth daily operations and efficient communication within the department.

2024 - 2025

Brgy. Mabato, Ormoc City

SK Chairperson

- Led the planning, implementation, and evaluation of various youth development programs, community outreach initiatives, and capacity-building activities in alignment with local government priorities.
- Oversaw the execution of youth-focused events including sports tournaments, clean-up drives, leadership training, and educational seminars.
- Maintained accurate documentation of projects, financial reports, and official resolutions for audit and compliance purposes.

Other Related Skills

- Photography
- Basic Graphic Designing (Canva)
- Video Creation

Seminars Attended

- SK Planning, Budgeting and Review Process Training Workshop
- Training on Foresight and Anticipatory Leadership in Local Governance and Development

Reference

ENGR. MA. LELENA TRINIDAD OJT Adviser

ACLC College of Ormoc City, Ormoc City

Phone: 0970 - 378 - 9767

MRS. ELLEN GUIÑARES

Adviser of Computer Studies Department

Phone: 0961 - 971 - 3200